
FORGE: Reaction-Guided Generative Design of Ionizable Lipids

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Ionizable lipids are essential for RNA delivery, but their discovery largely depends
2 on synthesizing and screening libraries assembled from predefined reactants. This
3 strategy preserves a clear synthesis route while limiting the chemical space that
4 can be explored. *De novo* generation could expand this space, yet whole-molecule
5 models do not retain the reaction information needed to assemble their designs.
6 We introduce FORGE (Flow-matched, Open-ended, Route-resolved Generation
7 and Exploration), which uses known reaction chemistry to design ionizable lipids
8 without restricting generation to a fixed reactant catalogue. Each reaction program
9 specifies precursor roles, reaction-core positions, transformation order and coarse
10 molecular architecture, while FORGE generates the complete molecule at atom-
11 and-bond resolution without component identifiers, stored precursor graphs or
12 fragment tokens. We evaluated one shared model on an amine–aldehyde–isocyanide
13 Ugi 3-CR, repeated aza-Michael addition and repeated reductive amination. Across
14 three independent seeds, verified exact level-one (exact-L1) yields were $96.4 \pm$
15 2.1% , $72.2 \pm 1.6\%$ and $52.3 \pm 3.4\%$, versus $34.8 \pm 7.1\%$, $19.2 \pm 4.1\%$ and $37.0 \pm$
16 2.2% for matched whole-molecule generation followed by post-hoc verification.
17 FORGE also generated exact-L1 products with precursor identities absent from
18 the corresponding training component catalogues across all three reactions, a
19 capability unavailable to finite-catalogue assemblers. These results establish that
20 known reaction chemistry can guide *de novo* molecular generation while preserving
21 the information needed to verify assembly.

1 Introduction

23 Lipid nanoparticles (LNPs) deliver nucleic-acid therapeutics including small interfering RNA, mRNA
24 vaccines, and *in vivo* genome-editing reagents [Adams et al., 2018, Polack et al., 2020, Baden et al.,
25 2021, Gillmore et al., 2021]. Within each LNP, the ionizable lipid supports nucleic-acid complexation,
26 particle assembly, and endosomal escape, making its chemical structure a major determinant of
27 delivery [Cullis and Hope, 2017, Hou et al., 2021]. Small changes to the headgroup or hydrophobic
28 domains can substantially alter function. Their effects depend on the complete molecular structure
29 and do not follow simple design rules [Cheng et al., 2020, Han et al., 2021, Hatit et al., 2023].
30 Researchers therefore discover ionizable lipids by defining modular precursor libraries, synthesizing
31 combinations, and screening the resulting LNPs experimentally [Akinc et al., 2008, Love et al., 2010,
32 Li et al., 2024, Xu et al., 2024]. This strategy preserves synthetic tractability and supports systematic
33 testing within defined chemical families, while limiting the search to components selected before
34 design begins.

35 *De novo* molecular generation has the potential to extend ionizable-lipid discovery beyond fixed
36 precursor libraries. Current synthesis-aware generators construct molecules by selecting reactants

and applying predefined transformations [Gao et al., 2022, Koziarski et al., 2024, Cretu et al., 2025, Rekesh et al., 2025]. This construction retains an explicit assembly route and limits generated molecules to the supplied reactant inventory. While whole-molecule generators can propose structures beyond a fixed inventory, their outputs do not retain the precursor roles and transformations needed to assemble each design [Qin et al., 2025, Lee et al., 2025]. Other approaches estimate synthesizability after generation without incorporating the intended assembly chemistry into the generative process [Gao and Coley, 2020]. Recent studies have also applied *de novo* generative models to ionizable lipids [Ou et al., 2024, Zhao et al., 2024]. The missing capability, however, is generation of complete molecular structures under known reaction chemistry while leaving precursor identities open.

To mitigate this, we introduce **FORGE** (**F**low-matched, **O**pen-ended, **R**oute-resolved **G**eneration and **E**xploration), a generative model that uses known reaction chemistry to design ionizable lipids beyond a fixed reactant catalogue. Each reaction program contains the assembly chemistry, precursor roles, reaction-core positions, transformation order, and coarse molecular architecture while leaving precursor identities unspecified. **FORGE** generates complete molecular graphs at atom-and-bond resolution without component identifiers, stored precursor graphs, or fragment tokens. We then decompose each completed graph into its implied precursors and require exact forward replay of the specified assembly reaction to reconstruct the generated molecule. For unavailable precursors, a bounded recursive search returns either a computational synthesis dossier supported by reaction and supplier evidence or an explicit unresolved or search-censored outcome.

We evaluate one shared **FORGE** model across an **AGILE**-type amine–aldehyde–isocyanide Ugi 3-CR, repeated aza-Michael addition, and repeated reductive amination [Xu et al., 2024, Li et al., 2023, Xue et al., 2024]. Across three independent seeds, **FORGE** achieves verified exact-L1 yields of $96.4 \pm 2.1\%$, $72.2 \pm 1.6\%$, and $52.3 \pm 3.4\%$, respectively, exceeding matched whole-molecule generation followed by post-hoc verification in every program. **FORGE** also produces verified exact assemblies with recovered precursor constitutions absent from the corresponding training catalogues. These experiments evaluate generalization across specified reaction programs and do not claim zero-shot support for arbitrary chemistry.

We summarize our main contributions as follows.

- **FORGE** generates complete molecular graphs using known reaction chemistry while leaving precursor identities open and does not select from stored reactants or fragment vocabularies.
- Matched controls across three assembly programs showed that reaction-program conditioning substantially increased exact transform-consistent generation relative to whole-molecule generation followed by post-hoc verification.
- Across all three programs, **FORGE** generated verified exact assemblies with recovered precursor constitutions absent from the corresponding training component catalogues. This establishes component-level expansion beyond the fixed training inventories; strong product-level catalogue escape remains unclaimed.

2 **FORGE**

FORGE generates complete ionizable-lipid graphs under specified assembly chemistry while leaving precursor identities open. Let $G \in \mathcal{G}$ denote a complete ionizable-lipid graph and $P \in \mathcal{P}$ a reaction program. **FORGE** models

$$G \sim p_{\theta}(\cdot | P) \quad (1)$$

without receiving precursor identifiers, stored component graphs or fragment tokens. Figure 1 summarizes reaction-program construction, discrete flow generation, exact assembly verification and optional route assessment.

2.1 **Reaction programs**

A reaction program is a tuple

$$P = (c, d, \mathbf{r}, \mathbf{k}, M), \quad (2)$$

where c identifies the ordered assembly chemistry, d is the program depth, $\mathbf{r}$ assigns precursor roles and $\mathbf{k}$ marks reaction-core positions. The morphology variable M specifies the exterior atom count, junction budget, cycle rank and attachment count for each precursor role. These values constrain

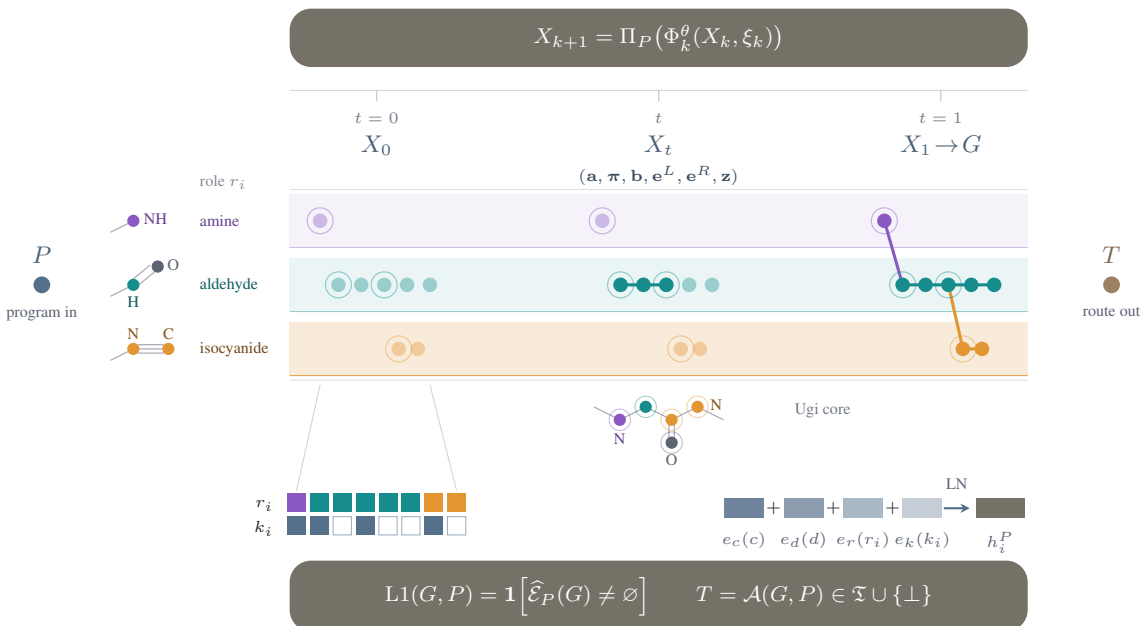


Figure 1: **FORGE generation and route assessment.** A reaction program fixes assembly roles and core coordinates without selecting precursor identities. FORGE generates the whole graph, verifies exact assembly replay, and recursively traces its implied precursors to evidence-qualified materials or an explicit abstention. Colours mark precursor origin; grey denotes the Ugi template-introduced oxygen. Replay and route closure are computational checks, not experimental synthesis evidence.

coarse molecular architecture while leaving the exact topology, atoms, bonds and precursor identities unspecified.

We sample an admissible complete program P from a training-fold program prior and generate G under the resulting program,

$$p_\theta(G, P | c) = p_{\text{prog}}(P | c) p_\theta(G | P). \quad (3)$$

At graph position i , the program embedding is

$$h_i^P = \text{LN} \left(e_c(c) + e_d(d) + e_r(r_i) + e_k(k_i) + \sum_{u=1}^4 e_{M,u}(M_{r_i,u}) \right). \quad (4)$$

This embedding identifies the position’s precursor role, reaction-core status and role-local morphology without encoding a component identity. The morphology fields also determine active positions and fixed or variable masks.

2.2 Discrete flow generation

FORGE represents a complete molecule using a categorical tree-plus-closure state. Given the selected exact atom-mapped annotation α , the deterministic program-aware encoder assigns

$$X_G := \text{Enc}_P(G, \alpha) = (\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}), \quad (5)$$

where $\mathbf{a}$ contains atom states, $\boldsymbol{\pi}$ contains parent pointers, $\mathbf{b}$ contains spanning-tree bond states and $(\mathbf{e}^L, \mathbf{e}^R, \mathbf{z})$ encodes closure bonds. The encoder fixes the root, vertex order, spanning tree and closure order as specified in Appendix E. The decoder is surjective onto $\mathfrak{G}_{N_{\text{max}}, K_{\text{cl}}}$, with $N_{\text{max}} = 194$ atoms and at most $K_{\text{cl}} = 3$ closure bonds. Every graph within the declared atom, bond and size support therefore has a valid representation, although representability does not imply reaction consistency or nonzero model probability; malformed raw states decode to $\perp_{\text{dec}}$.

Let $\mathcal{V}(P)$ and $\mathcal{F}(P)$ denote the variable and fixed coordinate sets. The program-compatible source fixes $\mathcal{F}(P)$ and factorizes across $\mathcal{V}(P)$. Given a clean endpoint $x_1 \in \Omega_P$, the conditional path is

$$q_{t|1}^P(x | x_1) = \mathbf{1}[x_{\mathcal{F}(P)} = x_{1,\mathcal{F}(P)}] \prod_{i \in \mathcal{V}(P)} [(1-t)q_{0,i}^P(x_i) + t\mathbf{1}\{x_i = x_{1,i}\}]. \quad (6)$$

Thus each variable coordinate is linearly interpolated while the joint conditional law is their product. A local continuous-time generator changes at most one coordinate infinitesimally, so the predicted clean-coordinate marginals determine its joint rate. Sampling begins from $X_0 \sim q_0^P$ and applies a capped independent-coordinate Euler approximation. Appendix E defines the rates and numerical kernel; the implementation does not claim exact continuous-time simulation.

FORGE applies this discrete flow to X_G and conditions endpoint prediction on P . A chemistry adapter converts the reaction program into masks over atom, pointer and bond coordinates. Program-fixed coordinates remain constant while the flow generates all remaining states. A graph Transformer predicts the clean atom, bond, pointer and closure states from (X_t, t, P) .

Let $\mathcal{J}$ denote the six state families, $m_j^{(q)}(P) \in \{0, 1\}$ indicate whether coordinate j in family q remains variable, and $\ell_j^{(q)}(\theta) = -\log p_\theta(X_{G,j}^{(q)} | X_t, t, P)$ denote the corresponding denoising loss. The primary masked objective is

$$\mathcal{L}_{\text{FORGE}}(\theta) = \mathbb{E}_{(G,P) \sim \mu_{\text{bal}}, t \sim \mathcal{U}(0,1), X_t \sim q_{t|1}^P(\cdot | X_G)} \left[\sum_{q \in \mathcal{J}} \lambda_q \frac{\sum_j m_j^{(q)}(P) \ell_j^{(q)}(\theta)}{\max\{1, \sum_j m_j^{(q)}(P)\}} \right], \quad (7)$$

where μ_{bal} assigns equal total training mass to each reaction program, λ_q weights family q , and a family with no variable coordinates contributes zero loss. Auxiliary role and core gradients are combined by the deterministic PCGrad procedure in Appendix B; the population result applies only to the displayed primary term.

During corruption and generation, fixed coordinates have zero update rate and the projection Π_P restores them after every finite step. Numerical sampling therefore follows

$$X_{k+1} = \Pi_P(\Phi_k^\theta(X_k, \xi_k)), \quad k = 0, \dots, L_{\text{samp}} - 1, \quad (8)$$

where Φ_k^θ is the capped Euler update and ξ_k contains its sampling randomness. This is the precise syntactic sense in which the program constrains the trajectory: program-fixed coordinates cannot be resampled. Chemical consistency with the requested assembly is established separately by exact forward replay. The terminal state is decoded into G and must be connected, valence-valid and chemically sanitized. Appendix E gives the representation and ideal population result, which excludes auxiliary losses, PCGrad, finite capacity and numerical integration error.

2.3 Assembly verification and route assessment

After generation, we apply the inverse transform associated with P to recover the implied precursors. The molecule satisfies verified exact level-one assembly only when the corresponding forward transform reconstructs G exactly. This test verifies the specified final reaction without requiring the generator to select from a stored precursor catalogue.

For precursors that are not directly available, a bounded program constructor searches for preparation routes and checks terminal supplier evidence,

$$Y = \text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T) \in \mathfrak{D} \cup \mathfrak{F}_{\mathfrak{B}}, \quad (9)$$

where $\mathfrak{D}$ is the dossier space, $\mathfrak{F}_{\mathfrak{B}}$ is the finite set of typed outcomes under the frozen route contract, and $(\mathbf{C}_T, \tau_T)$ is the sole verifier-returned exact root trace; ambiguous roots cause an abstention. A dossier is route-complete only when G passes molecular and exact-assembly verification, each upstream reaction is admitted and verified, and every required precursor traces to an available starting material. Otherwise the constructor returns an explicit unresolved or search-censored outcome.

Generation and route assessment remain separate. The reaction program P conditions the distribution of G , whereas $\text{Assess}_{\mathfrak{B}}$ evaluates the completed graph without modifying it. Appendix B provides the complete architecture, sampler, verifier and route-assessment procedure. Appendix E gives the theoretical results.

3 Results

We evaluated FORGE on Ugi, repeated aza-Michael addition and repeated reductive amination using three matched arms: shared conditioning, shared training without program information and

shared training with cyclically incorrect reaction labels. All arms used the same architecture, optimization budget, attempt count and prespecified final checkpoint. We evaluated three independent training seeds. Verified exact-L1 yield is the fraction of generation attempts that passed molecular validation, program-aware decomposition and exact forward replay. Component-disjoint held-out products entered neither training nor checkpoint selection, and generated molecules are not treated as independent training replicates. Appendix C provides the full protocol and seed-level results.

3.1 One model learns three assembly programs

We first tested whether one conditional flow could learn all three reactions while preserving Ugi performance. Across 3,072 held-out attempts per program and seed, FORGE achieved verified exact-L1 yields of $96.4 \pm 2.1\%$ for Ugi, $72.2 \pm 1.6\%$ for repeated aza-Michael addition and $52.3 \pm 3.4\%$ for repeated reductive amination (Table 1). Every verifier-returned trace passed exact forward replay in every program and seed, as required by construction. Reductive amination showed greater between-seed variation and higher decomposition ambiguity, indicating less consistent learning in this lower-data setting.

Shared conditioning increased Ugi exact-L1 yield from $34.8 \pm 7.1\%$ under shared-null training to $96.4 \pm 2.1\%$, a 61.6-point improvement (three-seed range 57.4 to 67.7 points). FORGE also produced no fixed-coordinate violations or support overflows across 27,648 held-out attempts. One shared flow therefore learned all three assembly programs while retaining the strongest Ugi behavior.

Table 1: **One shared conditional flow learns three ionizable-lipid assembly programs.** Exact-L1 yields are mean $\pm$ sample standard deviation across three seeds. The final column reports decomposition coverage and the forward-replay verifier invariant. Per-seed values: Appendix C.

Program	FORGE conditioned	Shared null	Cyclic program	Decomp. coverage / replay invariant (%)
Ugi	96.4 ± 2.1	34.8 ± 7.1	93.8 ± 4.2	99.9/100.0
Aza-Michael	72.2 ± 1.6	19.2 ± 4.1	71.5 ± 2.2	100.0/100.0
Reductive amination	52.3 ± 3.4	37.0 ± 2.2	50.7 ± 4.9	86.6/100.0

3.2 Reaction information changes generation

We next tested whether reaction information changed the generated distribution before verification. Under matched attempt budgets, conditioned generation achieved exact-L1 yields of $96.4 \pm 2.1\%$, $72.2 \pm 1.6\%$ and $52.3 \pm 3.4\%$ for Ugi, aza-Michael addition and reductive amination. Shared generation without program information followed by the same post-hoc verifier yielded $34.8 \pm 7.1\%$, $19.2 \pm 4.1\%$ and $37.0 \pm 2.2\%$.

Assigning the wrong reaction identity reduced exact-L1 yield to $93.8 \pm 4.2\%$, $71.5 \pm 2.2\%$ and $50.7 \pm 4.9\%$ across the three programs. Training without program information also reduced exact-L1 yield in every program. These controls show that the reaction program changes molecular generation before downstream verification. The small conditioned-versus-cyclic differences, however, do not identify a strong independent contribution from reaction identity alone.

3.3 Whole-graph generation recovers precursor constitutions outside training catalogues

We next asked whether FORGE could preserve exact assembly while generating precursor constitutions absent from training. Under the common Ugi benchmark, the finite-inventory methods produced 1000.0 ± 0.0 verified exact-L1 products per 1,000 attempts but zero component-novel products by construction. FORGE produced 963.9 ± 21.4 verified products and 862.6 ± 19.2 distinct component-novel products per 1,000 attempts (Table 2).

Table 2: **FORGE combines exact Ugi assembly with precursor constitutions absent from method-visible training.** Values are mean $\pm$ sample standard deviation per 1,000 attempts across three seeds under a common component-disjoint protocol and verifier. Component-novel is verifier-relative and does not establish strong product-level catalogue escape. Whole-molecule baselines are assessed post hoc. $\dagger$, structural zero; N/E, not estimable.

Models	Valid / 1,000 $\uparrow$	Verified L1 / 1,000 $\uparrow$	Distinct L1 / 1,000 $\uparrow$	Comp.-novel / 1,000 $\uparrow$	Held-comp. / 1,000 $\uparrow$	Diversity
RGFN	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0 †	0.0 $\pm$ 0.0	N/E
DeFoG unconditional	346.5 $\pm$ 42.3	280.6 $\pm$ 26.6	280.6 $\pm$ 26.6	280.6 $\pm$ 26.6	15.5 $\pm$ 2.4	0.727 $\pm$ 0.014
GenMol/SAFE	627.5 $\pm$ 6.2	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.873 $\pm$ 0.009
Learned inventory selector	1000.0 $\pm$ 0.0	1000.0 $\pm$ 0.0	849.5 $\pm$ 13.7	0.0 $\pm$ 0.0 †	0.0 $\pm$ 0.0	0.717 $\pm$ 0.001
Finite catalogue oracle	1000.0 $\pm$ 0.0	1000.0 $\pm$ 0.0	837.6 $\pm$ 8.2	0.0 $\pm$ 0.0 †	0.0 $\pm$ 0.0	0.719 $\pm$ 0.000
Shared-null + post-hoc	194.7 $\pm$ 15.6	114.4 $\pm$ 10.9	108.5 $\pm$ 9.7	108.5 $\pm$ 9.7	0.4 $\pm$ 0.2	0.754 $\pm$ 0.014
FACT-matched	60.8 $\pm$ 20.2	52.0 $\pm$ 15.6	50.8 $\pm$ 14.4	50.8 $\pm$ 14.4	0.0 $\pm$ 0.0	0.667 $\pm$ 0.009
FACT-generous	500.7 $\pm$ 213.8	296.7 $\pm$ 186.0	288.3 $\pm$ 175.3	288.3 $\pm$ 175.3	0.1 $\pm$ 0.2	0.774 $\pm$ 0.007
FORGE Transformer	965.0 $\pm$ 21.9	963.9 $\pm$ 21.4	862.6 $\pm$ 19.2	862.6 $\pm$ 19.2	1.4 $\pm$ 1.7	0.643 $\pm$ 0.022

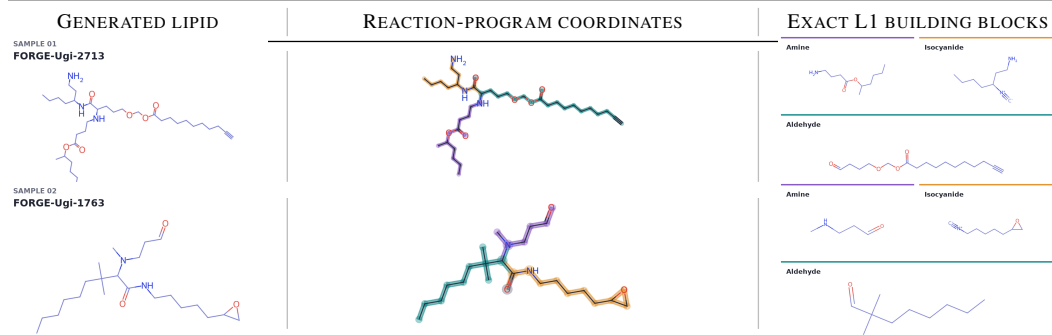
Against the fixed training catalogues, component-novel verified exact-L1 yields reached 839.2, 159.3 and 58.7 products per 1,000 attempts across Ugi, aza-Michael addition and reductive amination. The corresponding finite catalogues had zero component-novel yield by construction (Table 8). Representative Ugi products are shown in Figure 2.

Among the external baselines, DeFoG produced 280.6 ± 26.6 verified products and recovered held components more often than FORGE (15.5 ± 2.4 versus 1.4 ± 1.7 per 1,000 attempts). GenMol/SAFE produced no exact Ugi-L1 product, and RGFN emitted no admissible output. These comparisons show that reaction consistency, verifier-relative component novelty and held-component recovery are distinct properties; they do not establish strong product-level catalogue escape.

3.4 Recursive synthesis programs connect generated lipids to evidence-qualified materials

We finally asked whether verified final assemblies closed to complete computational synthesis dossiers. Exact-L1 verification establishes the final assembly reaction but not a complete route. Only 30 of 11,021 components in the frozen evidence union closed under the exact-source library. FORGE therefore produced 963.9 ± 21.4 verified exact-L1 products per 1,000 attempts but 0.0 ± 0.0 complete dossiers. In a route-blind sample of 256 products, 0 closed, 256 were unresolved and 0 were search-censored (Table 10). We report these outcomes as evidence closure, not as proof that unresolved products are unsynthesizable.

Figure 2: **FORGE generates exact Ugi products beyond its visible training inventory.** Each row contains a deterministic seed-0 product with a unique verified exact-L1 decomposition and a verifier-recovered precursor absent from training. Columns show the generated graph, atom-mapped origins and recovered precursors; halos mark the Ugi core and colours encode atom origin. These are structural examples, not evidence of activity or experimental synthesis success.



4 Discussion

In this work, we present FORGE, a generative model that uses known assembly chemistry during whole-molecule generation while leaving precursor identities open. Across an AGILE-type amine-aldehyde-isocyanide Ugi 3-CR, repeated aza-Michael addition and repeated reductive amination, conditioning increased verified exact-L1 yield relative to matched post-hoc verification and produced exact assemblies with verifier-recovered precursors absent from the training catalogues. These results show that reaction information can guide molecular generation without making a finite component inventory part of the generative state. Direct catalogue-product reachability was not computed, so strong product-level catalogue escape is not claimed.

Evidence was strongest for Ugi; the other families were lower-yield stress tests. This computational study covers three specified programs within bounded graph support. Exact replay verifies the final assembly, not precursor availability, synthesis success, lipid activity or RNA delivery. Negative results include weak held-component recovery (1.4 ± 1.7 products per 1,000 attempts), failure to reproduce the held-out lipid distribution, no uniform advantage for joint denoising and no matched improvement from synthesis or potency guidance (Appendix C). Prospective synthesis, formulation and biological validation are outside scope. Within these limits, reaction programs expand molecular identity while retaining a verifiable connection to assembly chemistry.

References

- David Adams, Alejandra Gonzalez-Duarte, William D. O’Riordan, Chih-Chao Yang, Mitsuharu Ueda, Arnt V. Kristen, Ivailo Tournev, Hartmut H. Schmidt, Teresa Coelho, John L. Berk, Kon-Ping Lin, Giuseppe Vita, Shahram Attarian, Violaine Planté-Bordeneuve, Michelle M. Mezei, Josep M. Campistol, Juan Buades, Thomas H. Brannagan, Byoung J. Kim, Jeeyoung Oh, Yesim Parman, Yoshiki Sekijima, Philip N. Hawkins, Scott D. Solomon, Michael Polydefkis, Peter J. Dyck, Pritesh J. Gandhi, Sunita Goyal, Jihong Chen, Andrew L. Strahs, Saraswathy V. Nochur, Marianne T. Sweetser, Pushkal P. Garg, Akshay K. Vaishnav, Jared A. Gollob, and Ole B. Suhr. Patisiran, an RNAi therapeutic, for hereditary transthyretin amyloidosis. *New England Journal of Medicine*, 379(1):11–21, 2018. doi: 10.1056/NEJMoa1716153.
- Akin Akinc, Andreas Zumbuehl, Michael Goldberg, Elizaveta S. Leshchiner, Valentina Busini, Naushad Hossain, Sergio A. Bacallado, David N. Nguyen, Jason Fuller, Rene Alvarez, Anna Borodovsky, Todd Borland, Rainer Constien, Antonin de Fougères, J. Robert Dorkin, K. Narayanannair Jayaprakash, Muthusamy Jayaraman, Matthias John, Victor Kotliansky, Muthiah Manoharan, Lubomir Nechev, June Qin, Timothy Racie, Denitza Raitcheva, Kallanthottathil G. Rajeev, Dinah W. Y. Sah, Jürgen Soutschek, Ivanka Toudjarska, Hans-Peter Vornlocher, Tracy S. Zimmermann, Robert Langer, and Daniel G. Anderson. A combinatorial library of lipid-like materials for delivery of RNAi therapeutics. *Nature Biotechnology*, 26(5):561–569, 2008. doi: 10.1038/nbt1402.
- Lindsey R. Baden, Hana M. El Sahly, Brandon Essink, Karen Kotloff, Sharon Frey, Rick Novak, David Diemert, Stephen A. Spector, Nadine Rouphael, C. Buddy Creech, John McGittigan, Shishir Khetan, Nathan Segall, Joel Solis, Adam Brosz, Carlos Fierro, Howard Schwartz, Kathleen Neuzil, Lawrence Corey, Peter Gilbert, Holly Janes, Dean Follmann, Mary Marovich, John Mascola, Laura Polakowski, Julie Ledgerwood, Barney S. Graham, Hamilton Bennett, Rolando Pajon, Conor Knightly, Brett Leav, Weiping Deng, Honghong Zhou, Shu Han, Melanie Ivarsson, Jacqueline Miller, and Tal Zaks. Efficacy and safety of the mRNA-1273 SARS-CoV-2 vaccine. *New England Journal of Medicine*, 384(5):403–416, 2021. doi: 10.1056/NEJMoa2035389.
- Qiang Cheng, Tuo Wei, Lukas Farbiak, Lindsay T. Johnson, Sean A. Dilliard, and Daniel J. Siegwart. Selective organ targeting (SORT) nanoparticles for tissue-specific mRNA delivery and CRISPR-Cas gene editing. *Nature Nanotechnology*, 15(4):313–320, 2020. doi: 10.1038/s41565-020-0669-6.
- Miruna Cretu, Charles Harris, Iliia Igashov, Arne Schneuing, Marwin Segler, Bruno Correia, Julien Roy, Emmanuel Bengio, and Pietro Liò. SynFlowNet: Design of diverse and novel molecules with synthesis constraints. In *International Conference on Learning Representations*, 2025.
- Pieter R. Cullis and Michael J. Hope. Lipid nanoparticle systems for enabling gene therapies. *Molecular Therapy*, 25(7):1467–1475, 2017. doi: 10.1016/j.ymthe.2017.03.013.

- Wenhao Gao and Connor W. Coley. The synthesizability of molecules proposed by generative models. *Journal of Chemical Information and Modeling*, 60(12):5714–5723, 2020. doi: 10.1021/acs.jcim.0c00174.
- Wenhao Gao, Rocío Mercado, and Connor W. Coley. Amortized tree generation for bottom-up synthesis planning and synthesizable molecular design. In *International Conference on Learning Representations*, 2022.
- Samuel Genheden, Amol Thakkar, Veronika Chadimová, Jean-Louis Reymond, Ola Engkvist, and Esben Bjerrum. AiZynthFinder: A fast, robust and flexible open-source software for retrosynthetic planning. *Journal of Cheminformatics*, 12:70, 2020. doi: 10.1186/s13321-020-00472-1.
- Julian D. Gillmore, Ed Gane, Jorg Taubel, Justin Kao, Marianna Fontana, Michael L. Maitland, Jessica Seitzer, Daniel O’Connell, Kathryn R. Walsh, Kristy Wood, Jonathan Phillips, Yuanxin Xu, Adam Amaral, Adam P. Boyd, Jeffrey E. Cehelsky, Mark D. McKee, Andrew Schiermeier, Olivier Harari, Andrew Murphy, Christos A. Kyratsous, Brian Zambrowicz, Randy Soltys, David E. Gutstein, John Leonard, Laura Sepp-Lorenzino, and David Lebowitz. CRISPR-Cas9 in vivo gene editing for transthyretin amyloidosis. *New England Journal of Medicine*, 385(6):493–502, 2021. doi: 10.1056/NEJMoa2107454.
- Xuexiang Han, Hanwen Zhang, Kamila Butowska, Kelsey L. Swingle, Mohamad-Gabriel Alameh, Drew Weissman, and Michael J. Mitchell. An ionizable lipid toolbox for RNA delivery. *Nature Communications*, 12:7233, 2021. doi: 10.1038/s41467-021-27493-0.
- Marine Z. C. Hatit, Curtis N. Dobrowolski, Melissa P. Lokugamage, David Loughrey, Huanzhen Ni, Chiara Zurla, Alejandro J. Da Silva Sanchez, Afsane Radmand, Sebastian G. Huayamares, Ryan Zenhausern, Kalina Paunovska, Hannah E. Peck, Jinwhan Kim, Manaka Sato, Jacob I. Feldman, Michael-Alexander Rivera, Ana Cristian, YongTae Kim, Philip J. Santangelo, and James E. Dahlman. Nanoparticle stereochemistry-dependent endocytic processing improves in vivo mRNA delivery. *Nature Chemistry*, 15(4):508–515, 2023. doi: 10.1038/s41557-023-01138-9.
- Xucheng Hou, Tal Zaks, Robert Langer, and Yizhou Dong. Lipid nanoparticles for mRNA delivery. *Nature Reviews Materials*, 6(12):1078–1094, 2021. doi: 10.1038/s41578-021-00358-0.
- Muthusamy Jayaraman, Steven M. Ansell, Barbara L. Mui, Ying K. Tam, Jianxin Chen, Xinyao Du, David Butler, Laxman Eltepu, Shigeo Matsuda, Jayaprakash K. Narayanannair, Kallanthottathil G. Rajeev, Ismail M. Hafez, Akin Akinc, Martin A. Maier, Mark A. Tracy, Pieter R. Cullis, Thomas D. Madden, Muthiah Manoharan, and Michael J. Hope. Maximizing the potency of siRNA lipid nanoparticles for hepatic gene silencing in vivo. *Angewandte Chemie International Edition*, 51(34):8529–8533, 2012. doi: 10.1002/anie.201203263.
- Michał Koziarski, Andrei Rekesh, Dmytro Shevchuk, Almer van der Sloot, Piotr Gaiński, Yoshua Bengio, Cheng-Hao Liu, Mike Tyers, and Robert Batey. RGFN: Synthesizable molecular generation using GFlowNets. In *Advances in Neural Information Processing Systems*, volume 37, pages 46908–46955, 2024.
- Seul Lee, Karsten Kreis, Srimukh Prasad Veccham, Meng Liu, Danny Reidenbach, Yuxing Peng, Saeed Gopal Paliwal, Weili Nie, and Arash Vahdat. GenMol: A drug discovery generalist with discrete diffusion. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 33205–33226. PMLR, 2025.
- Bowen Li, Rajith Singh Manan, Shun-Qing Liang, Akiva Gordon, Allen Jiang, Andrew Varley, Guangping Gao, Robert Langer, Wen Xue, and Daniel Anderson. Combinatorial design of nanoparticles for pulmonary mRNA delivery and genome editing. *Nature Biotechnology*, 41(10):1410–1415, 2023. doi: 10.1038/s41587-023-01679-x.
- Bowen Li, Idris O. Raji, Akiva G. R. Gordon, Lizhuang Sun, Theresa M. Raimondo, Favour A. Oladimeji, Allen Y. Jiang, Andrew Varley, Robert S. Langer, and Daniel G. Anderson. Accelerating ionizable lipid discovery for mRNA delivery using machine learning and combinatorial chemistry. *Nature Materials*, 23(7):1002–1008, 2024. doi: 10.1038/s41563-024-01867-3.

Kevin T. Love, Kerry P. Mahon, Christopher G. Levins, Kathryn A. Whitehead, William Querbes, J. Robert Dorkin, June Qin, William Cantley, Liu Liang Qin, Timothy Racie, Maria Frank-Kamenetsky, Ka Ning Yip, Rene Alvarez, Dinah W. Y. Sah, Antonin de Fougères, Kevin Fitzgerald, Victor Kotliansky, Akin Akinc, Robert Langer, and Daniel G. Anderson. Lipid-like materials for low-dose, in vivo gene silencing. *Proceedings of the National Academy of Sciences*, 107(5):1864–1869, 2010. doi: 10.1073/pnas.0910603106.

Yuxuan Ou, Jingyi Zhao, Austin Tripp, Morteza Rasoulboroujeni, and José Miguel Hernández-Lobato. A deep generative model for the design of synthesizable ionizable lipids. In *NeurIPS 2024 Workshop on AI for New Drug Modalities*, 2024. arXiv:2412.00928.

Fernando P. Polack, Stephen J. Thomas, Nicholas Kitchin, Judith Absalon, Alejandra Gurtman, Stephen Lockhart, John L. Perez, Gonzalo Pérez Marc, Edson D. Moreira, Cristiano Zerbini, Ruth Bailey, Kena A. Swanson, Satrajit Roychoudhury, Kenneth Koury, Ping Li, Warren V. Kalina, David Cooper, Robert W. Frenck, Laura L. Hammitt, Özlem Türeci, Haylene Nell, Axel Schaefer, Serhat Ünal, Dina B. Tresnan, Susan Mather, Philip R. Dormitzer, Uğur Şahin, Kathrin U. Jansen, and William C. Gruber. Safety and efficacy of the BNT162b2 mRNA Covid-19 vaccine. *New England Journal of Medicine*, 383(27):2603–2615, 2020. doi: 10.1056/NEJMoa2034577.

Yiming Qin, Manuel Madeira, Dorina Thanou, and Pascal Frossard. DeFoG: Discrete flow matching for graph generation. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 50269–50326. PMLR, 2025.

Andrei Reakesh, Miruna Cretu, Dmytro Shevchuk, Vignesh Ram Somnath, Pietro Liò, Robert A. Batey, Mike Tyers, Michał Koziarski, and Cheng-Hao Liu. SynCoGen: Synthesizable 3d molecule generation via joint reaction and coordinate modeling. arXiv:2507.11818, 2025.

Sean C. Semple, Akin Akinc, Jianxin Chen, Ammen P. Sandhu, Barbara L. Mui, Connie K. Cho, Dinah W. Y. Sah, Derrick Stebbing, Erin J. Crosley, Ed Yaworski, Ismail M. Hafez, J. Robert Dorkin, June Qin, Kieu Lam, Kallanthottathil G. Rajeev, Kim F. Wong, Lloyd B. Jeffs, Lubomir Nechev, Merete L. Eisenhardt, Muthusamy Jayaraman, Mikameh Kazem, Martin A. Maier, Masuna Srinivasulu, Michael J. Weinstein, Qingmin Chen, Rene Alvarez, Scott A. Barros, Soma De, Sandra K. Klimuk, Todd Borland, Verbena Kosovrasti, William L. Cantley, Ying K. Tam, Muthiah Manoharan, Marco A. Ciufolini, Mark A. Tracy, Antonin de Fougères, Ian MacLachlan, Pieter R. Cullis, Thomas D. Madden, and Michael J. Hope. Rational design of cationic lipids for siRNA delivery. *Nature Biotechnology*, 28(2):172–176, 2010. doi: 10.1038/nbt.1602.

Yue Xu, Shihao Ma, Haotian Cui, Jingan Chen, Shufen Xu, Fanglin Gong, Alex Golubovic, Muye Zhou, Kevin Chang Wang, Andrew Varley, Rick Xing Ze Lu, Bo Wang, and Bowen Li. AGILE platform: A deep learning powered approach to accelerate LNP development for mRNA delivery. *Nature Communications*, 15:6305, 2024. doi: 10.1038/s41467-024-50619-z.

Lulu Xue, Alex G. Hamilton, Gan Zhao, Zebin Xiao, Rakan El-Mayta, Xuexiang Han, Ningqiang Gong, Xinhong Xiong, Junchao Xu, Christian G. Figueroa-Espada, Sarah J. Shepherd, Alvin J. Mukalel, Mohamad-Gabriel Alameh, Jiaxi Cui, Karin Wang, Andrew E. Vaughan, Drew Weissman, and Michael J. Mitchell. High-throughput barcoding of nanoparticles identifies cationic, degradable lipid-like materials for mRNA delivery to the lungs in female preclinical models. *Nature Communications*, 15(1):1884, 2024. doi: 10.1038/s41467-024-45422-9.

Jingyi Zhao, Yuxuan Ou, Austin Tripp, Morteza Rasoulboroujeni, and José Miguel Hernández-Lobato. Generative model for synthesizing ionizable lipids: A Monte Carlo tree search approach. arXiv:2412.00807, 2024.

A Preliminaries

Appendices A through F are supplementary and are excluded from the six-page content limit, as are the references. They give the full formulation, the complete reporting contract, the theoretical results and a generated-structure atlas.

A.1 Ionizable-lipid structure and modular assembly

The structure of an ionizable lipid is typically organized into three functional regions: a protonatable headgroup, a linker, and one or more hydrophobic domains [Semple et al., 2010, Jayaraman et al., 2012]. Together, these regions determine the ionization, stability, lipid packing, and membrane interactions of the molecule. Ionizable-lipid libraries are often synthesized by applying a common reaction to different combinations of precursor molecules [Akinc et al., 2008, Love et al., 2010, Li et al., 2024, Xu et al., 2024]. Each precursor contributes a defined portion of the final lipid and contains the reactive group required for the assembly. The reaction forms the bonds that connect these precursor-derived regions. As a result, the assembly reaction determines how the regions are connected, while the precursor structures determine the molecular identity of the resulting lipid.

A.2 Assembly chemistries studied

We study three ionizable-lipid assembly chemistries: an AGILE-type amine–aldehyde–isocyanide Ugi three-component reaction, repeated aza-Michael addition, and repeated reductive amination. In the AGILE-type Ugi reaction, an amine head reacts with an aldehyde component and an isocyanide component to form an α -amino amide that connects the headgroup to two hydrophobic regions [Xu et al., 2024]. A second assembly strategy uses repeated aza-Michael additions to attach acrylate-bearing hydrophobic components sequentially to reactive amines in the headgroup [Li et al., 2023]. Reductive amination sequentially couples aldehyde-bearing hydrophobic components to reactive amines in the headgroup through new carbon–nitrogen bonds [Xue et al., 2024]. Figure 3 draws all three.

A.3 Reaction transforms and exact assembly

Each assembly chemistry defines a forward transformation that maps role-assigned precursor molecules to the product graphs permitted by that reaction. The corresponding reverse transformation decomposes a complete lipid into the precursor assignments that could have produced it. To verify a decomposition, we computationally replay the reaction on the recovered precursors. The product passes the level-one (L1) assembly check only if the reconstruction matches the original lipid exactly.

A.4 Discrete molecular graph spaces

Let Σ_{atom} and Σ_{bond} be finite atom-state and bond-state vocabularies. A molecular graph is $G = (V, E, \ell_V, \ell_E)$, where the underlying graph is finite, connected, undirected and simple, $\ell_V : V \rightarrow \Sigma_{\text{atom}}$, and $\ell_E : E \rightarrow \Sigma_{\text{bond}}$. Bond order is part of ℓ_E rather than a parallel edge. Graphs are identified up to label-preserving isomorphism. The declared model support additionally bounds the heavy-atom count and cycle rank.

A.5 Discrete flow matching

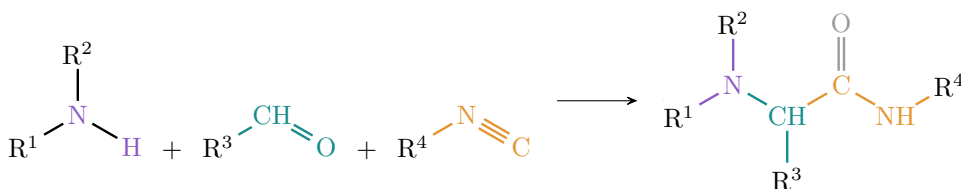
For one coordinate with alphabet $\mathcal{X}_i$, full-support source marginal $q_{0,i}$ and clean endpoint $a \in \mathcal{X}_i$, the linear conditional path is

$$q_{t|1,i}(u | a) = (1 - t)q_{0,i}(u) + t \mathbf{1}\{u = a\}, \quad t \in [0, 1], \quad (10)$$

with a univariate conditional rate $R_{t,i}^*(u, v | a)$ that realizes this path [Qin et al., 2025]. FORGE applies this construction independently to the variable coordinates conditional on the shared clean endpoint. Consequently the joint conditional path is a product of coordinate paths, not the mixture $(1 - t)q_0 + tq_1$. Its continuous-time generator permits one coordinate jump at an infinitesimal time. The denoiser predicts the clean posterior marginal of each coordinate, which is exactly the quantity needed to marginalize the corresponding local rate. Appendix B defines the full path, learned generator and the capped independent-coordinate Euler approximation used in code.

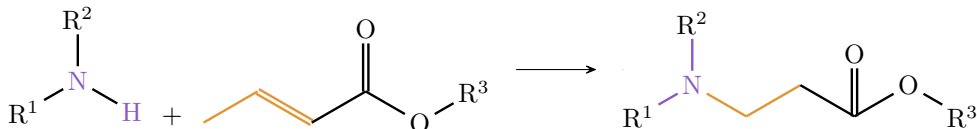
Ugi three-component reaction

amine head, aldehyde and isocyanide combine in one step; no carboxylic-acid reactant component



Repeated aza-Michael addition

the head N-H adds across the acrylate double bond; the ester is untouched



Repeated reductive amination

one new carbon-nitrogen bond; the carbonyl oxygen leaves as water

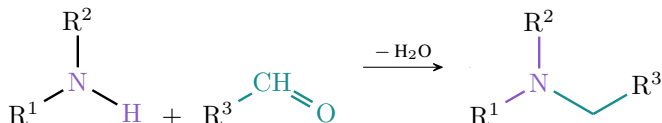


Figure 3: **The three assembly chemistries.** Skeletal notation, with each bond carrying the colour of the component that contributes it: amine head in purple, aldehyde-derived atoms in teal, the isocyanide or acrylate tail in orange. Grey marks an atom the assembly template introduces and no precursor supplies, following the same convention as the core inset of Figure 1. Hydrogens are drawn only on the atoms whose substitution decides whether a product can be decomposed back into precursors: the head N-H, and the α -carbon C-H that the Ugi reaction takes from the aldehyde. R^i stand for the source-linked components of the frozen library, which are larger than the fragments drawn here. The Ugi variant has no carboxylic-acid reactant component.

B FORGE in full

This appendix gives the complete formulation summarized in Section 2 of the body.

B.1 Overview

We formulate ionizable-lipid design as conditional generation under a specified reaction program, while leaving precursor identities unspecified. Let $G \in \mathcal{G}$ denote a complete ionizable-lipid graph and $P \in \mathcal{P}$ a reaction program specifying the assembly chemistry, precursor roles, reaction-core positions, and transformation order. FORGE models the conditional distribution $p_\theta(G | P)$ without access to precursor identifiers, stored component graphs, or fragment tokens. FORGE is organized into four stages:

- 1. Reaction-program specification.** P defines the assembly chemistry, precursor roles, reaction-core positions, transformation order, and coarse per-role morphology, but not precursor identities.
- 2. Whole-graph generation.** A conditional discrete flow generates every atom, bond, and topology state of G under P .
- 3. Exact assembly verification.** A reverse transform decomposes G into role-assigned precursors, and the corresponding forward transform must reconstruct G exactly.
- 4. Recursive synthesis-program construction.** FORGE searches for preparation routes to generated precursors that are not already supported as available starting materials and abstains when a route cannot be completed.

B.2 Synthesis-grounded design task

Given a reaction program P , FORGE must generate a complete molecular graph G and either construct a complete synthesis program for G or abstain. Let $\mathcal{D}$ denote the dossier space and let $\mathfrak{F}_{\mathfrak{B}}$ be the finite set of typed failure outcomes under a frozen route contract $\mathfrak{B}_{\text{route}}$. We write the design process as

$$G \sim p_\theta(\cdot | P), \quad Y = \text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T) \in \mathcal{D} \cup \mathfrak{F}_{\mathfrak{B}}. \quad (11)$$

$\text{Assess}_{\mathfrak{B}}$ denotes the bounded program constructor, invoked after the verifier selects the sole admitted root trace $(\mathbf{C}_T, \tau_T) \in \widehat{\mathcal{E}}_P(G)$, that verifies the final assembly, searches for preparation routes to unavailable precursors, and checks terminal supplier evidence. P and the assessor act at different stages: P guides generation, whereas the assessor constructs a dossier from the completed graph without modifying it. FORGE accepts (G, T) only when G passes molecular and exact-assembly verification and T traces every required precursor to an available starting material; otherwise, it abstains.

B.3 Reaction-program specification

Definition B.1 (Reaction program). *A reaction program is a tuple*

$$P = (c, d, \mathbf{r}, \mathbf{k}, M), \quad (12)$$

with an ordered layout index set $\mathcal{I}(P) = \{1, \dots, n(P)\}$. Here c identifies the ordered assembly chemistry, d is the program depth, $\mathbf{r} = (r_i)_{i \in \mathcal{I}(P)}$ assigns precursor roles, $\mathbf{k} = (k_i)_{i \in \mathcal{I}(P)}$ marks reaction-core positions, and M specifies coarse per-role morphology. For each precursor role r , M records four integers: the number of exterior atoms n_r , junction budget j_r , cycle rank γ_r , and attachment count a_r .

M is defined relative to an exact atom-mapped trace τ and precursor tuple $\mathbf{C}$. An atom is exterior for role r when τ assigns it to the corresponding precursor and does not mark it as reaction core. Template-introduced atoms belong to the core and to no precursor role. The role-induced subgraph contains the atoms assigned to r and all product bonds with both endpoints in that set. Its junction count is $\sum_v \max\{\deg_r(v) - 2, 0\}$, its cycle rank is $|E_r| - |V_r| + \kappa_r$ for κ_r connected components, and its core-attachment count is the number of product bonds between an exterior atom of r and a core

atom. We write $(G, \mathbf{C}, \tau) \models M(P)$ when these values match the sampled counts, with junction count at most its budget. Thus morphology is decomposition-relative rather than an unlabeled property of G .

FORGE first samples an admissible complete program P from the training-fold program prior and then generates G conditioned on it:

$$p_\theta(G, P | c) = p_{\text{prog}}(P | c) p_\theta(G | P). \quad (13)$$

The prior sets each region’s atom count, branching budget, cycle rank, and number of core attachments, but not its molecular structure. A chemistry-specific adapter defines the precursor roles, reaction core, reverse decomposition, and forward transform associated with c . For each graph position i , FORGE combines learned embeddings of the program identity, program depth, precursor role, and reaction-core position:

$$h_i^P = \text{LN} \left(e_c(c) + e_d(d) + e_r(r_i) + e_k(k_i) + \sum_{u=1}^4 e_{M,u}(M_{r_i,u}) \right). \quad (14)$$

The morphology fields also determine active positions and fixed/variable masks in the sampling layout. These coordinates tell the generator which precursor role each position plays, whether it lies in the reaction core, and the role-local count constraints, but not which molecular structure should occupy that role.

B.4 Program-defined molecular space

Reaction programs constrain how a lipid is assembled without specifying the molecular structures that occupy its precursor roles. Let $\mathcal{G}_{\text{bounded}} \subseteq \mathcal{G}$ denote the molecular graphs within the declared atom and bond vocabularies, the 194-heavy-atom limit, and the three-closure limit. For chemistry c , let $\mathcal{D}_c(G)$ be the candidate exact-decomposition search space and define

$$\mathcal{D}_P(G) = \{(\mathbf{C}, \tau) \in \mathcal{D}_c(G) : (G, \mathbf{C}, \tau) \models P\}. \quad (15)$$

The valid, exact-L1 molecular space associated with P is

$$\mathcal{G}_{\text{L1}}(P) = \left\{ G \in \mathcal{G}_{\text{bounded}} : \text{Valid}(G) = 1, \widehat{\text{L1}}(G, P) = 1 \right\}. \quad (16)$$

A generated lipid can pass the exact L1 check even when one or more recovered precursors never appeared in the training component set, provided the specified reaction reconstructs the lipid from them. For a reaction program P , let $\mathfrak{C}_P$ denote its admissible precursor-program inputs, including precursor identities, role assignments, and program depth. The registered forward operator maps each input to a finite set of canonical products:

$$\mathcal{F}_P : \mathfrak{C}_P \longrightarrow 2^{\mathfrak{G}^{N, K_{\text{cl}}}}. \quad (17)$$

Given role-specific catalogues $\mathcal{C}$, let $\mathfrak{C}_P(\mathcal{C}) \subseteq \mathfrak{C}_P$ contain the admissible program inputs whose components all belong to their corresponding catalogues. The catalogue-reachable product set is

$$\mathcal{G}_{\text{cat}}(P, \mathcal{C}) = \bigcup_{\mathbf{C} \in \mathfrak{C}_P(\mathcal{C})} \mathcal{F}_P(\mathbf{C}). \quad (18)$$

For any graph G , define its complete program-relative preimage as

$$\mathcal{E}_P^*(G) = \{\mathbf{C} \in \mathfrak{C}_P : G \in \mathcal{F}_P(\mathbf{C})\}. \quad (19)$$

Definition B.2 (Strong catalogue escape).

$$\text{Escape}^*(G; P, \mathcal{C}) = \mathbf{1}[\mathcal{E}_P^*(G) \neq \emptyset \wedge \mathcal{E}_P^*(G) \cap \mathfrak{C}_P(\mathcal{C}) = \emptyset]. \quad (20)$$

Proposition E.3 bounds the catalogue-reachable product set. For a verified exact-L1 graph, direct membership in the canonical product-hash set gives the operational strong-escape test

$$\text{StrongEscape}(G; P, \mathcal{C}) = \mathbf{1}[\widehat{\text{L1}}(G, P) = 1 \wedge G \notin \mathcal{G}_{\text{cat}}(P, \mathcal{C})]. \quad (21)$$

461 The verifier-returned set $\widehat{\mathcal{E}}_P(G)$ defined below is a subset of $\mathcal{E}_P^*(G)$ when every returned trace passes
 462 exact forward replay. A novel component in a returned trace establishes strong catalogue escape
 463 only when reverse enumeration is exhaustive or direct catalogue reachability is checked indepen-
 464 dently. The present experiments did not precompute a complete common-reference product-hash
 465 set, so they report verifier-recovered component novelty as a secondary diagnostic and do not report
 466 StrongEscape for FORGE. FORGE instead parameterizes $p_\theta(G | P)$ over whole graphs in $\mathcal{G}_{\text{bounded}}$,
 467 without including any component catalogue $\mathcal{C}_r$ in the generative state. Reaction programs constrain
 468 assembly semantics, while the flow determines precursor identity at atom and bond resolution.

469 B.5 Whole-graph generation

470 FORGE realizes this whole-graph distribution with a discrete flow over atom, bond, and topology
 471 states. To preserve the full 194-atom support without an $O(N_{\text{max}}^2)$ edge state, FORGE encodes
 472 each graph as a spanning tree plus at most three residual closure edges. Let $\alpha = (\mathbf{C}, \tau, \mathbf{r}, \mathbf{k})$
 473 contain the selected exact precursor tuple, atom-mapped reaction trace, precursor-role assignment
 474 and reaction-core annotation. The deterministic program-aware encoder

$$\text{Enc}_P : \{(G, \alpha) : G \in \mathfrak{G}_{N_{\text{max}}, K_{\text{cl}}} \text{ and } \alpha \text{ is admitted for } P\} \longrightarrow \mathfrak{X}_{N_{\text{max}}, K_{\text{cl}}} \quad (22)$$

475 constructs the categorical graph state

$$X_G := \text{Enc}_P(G, \alpha) = (\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}) \quad (23)$$

476 where $\mathbf{a}$ contains atom states, $\boldsymbol{\pi}$ parent pointers, $\mathbf{b}$ spanning-tree bond states, $\mathbf{e}^L$ and $\mathbf{e}^R$ closure-
 477 endpoint pointers, and $\mathbf{z}$ closure-bond states.

478 The encoder first partitions the atoms into connected precursor-origin components under α . It
 479 chooses as root the reaction-core atom with the smallest tied-broken canonical RDKit rank, and
 480 orders the origin-component graph by breadth-first traversal from that root component, breaking
 481 ties by role index, component canonical ranks and original atom index. Within each component it
 482 uses a deterministic depth-first preorder from the selected attachment atom, with neighbors ordered
 483 by the same canonical ranks. The traversed bonds define the spanning tree. Every residual bond
 484 is represented once as an ordered endpoint pair with its bond label, and residual triples are sorted
 485 lexicographically. Role, core, fixed-coordinate and morphology arrays are permuted by precisely the
 486 same atom order. Training records with more than one qualified exact source trace are not resolved
 487 arbitrarily: they abstain from semantic training; duplicate descriptions of the same admitted trace
 488 collapse to its canonical record. Thus $\text{Dec}(\text{Enc}_P(G, \alpha)) \cong G$ for every admitted record.

489 The raw padded categorical product space $\overline{\mathfrak{X}}_{N_{\text{max}}, K_{\text{cl}}}$ also contains malformed states. Its well-
 490 formed subset $\mathfrak{X}_{N_{\text{max}}, K_{\text{cl}}}$ enforces an active node prefix, earlier-parent pointers, distinct valid closure
 491 endpoints, no duplicate edges, and null values in unused closure slots. The partial decoder is therefore
 492 typed as

$$\text{Dec} : \overline{\mathfrak{X}}_{N_{\text{max}}, K_{\text{cl}}} \longrightarrow \mathfrak{G}_{N_{\text{max}}, K_{\text{cl}}} \cup \{\perp_{\text{dec}}\}. \quad (24)$$

493 For this study, $N_{\text{max}} = 194$ and $K_{\text{cl}} = 3$. This is a statement about architectural support. It does
 494 not imply reaction consistency, nonzero probability under a trained model, or synthesis-program
 495 completeness.

496 If $q_\theta(x | P)$ is the terminal raw-state law, the generated graph distribution is its decoder pushforward,

$$p_\theta([G]_{\cong} | P) = \sum_{x: \text{Dec}(x) \cong G} q_\theta(x | P), \quad p_\theta(\perp_{\text{dec}} | P) = \sum_{x: \text{Dec}(x) = \perp_{\text{dec}}} q_\theta(x | P). \quad (25)$$

497 The chemistry adapter marks any atom, pointer, or bond states fixed by P . FORGE holds these
 498 states constant while corrupting the remaining states along the linear probability path and predicting
 499 their clean values from (X_t, t, P) . Let Ω_P be the face of the raw state space satisfying those fixed
 500 coordinates, and define the admissible program set

$$\mathcal{P}_{\text{adm}} = \{P : \mathfrak{X}_{N_{\text{max}}, K_{\text{cl}}} \cap \Omega_P \neq \emptyset\}. \quad (26)$$

501 The morphology prior is normalized only on $\mathcal{P}_{\text{adm}}$. Each chemistry adapter is required to satisfy the
 502 program-compatible encoding property: every admitted annotated record obeys $\text{Enc}_P(G, \alpha) \in \Omega_P$.
 503 Corpus construction checks this invariant together with exact constitutional decode round trip and

fails closed before a record enters any split. This is an exhaustive implementation invariant over the cached training, calibration and held-out records, not a theorem for unregistered chemistries.

Let $\mathcal{J} = \{\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}\}$ denote the six state families optimized during training. Let μ_{bal} denote the frozen training distribution over graph-program pairs, which assigns equal total mass to each reaction program and preserves the source-balanced weights within each program. For each $q \in \mathcal{J}$, let $m_j^{(q)}(P) \in \{0, 1\}$ indicate whether coordinate j remains variable under P . FORGE minimizes a program-balanced denoising objective over the variable coordinates:

$$\mathcal{L}_{\text{FORGE}}(\theta) = \mathbb{E}_{(G,P) \sim \mu_{\text{bal}}, t \sim \mathcal{U}(0,1), X_t \sim p_t(\cdot | X_G, P)} \left[\sum_{q \in \mathcal{J}} \lambda_q \frac{\sum_j m_j^{(q)}(P) \left[-\log p_\theta \left(X_{G,j}^{(q)} | X_t, t, P \right) \right]}{\max \left\{ 1, \sum_j m_j^{(q)}(P) \right\}} \right]. \quad (27)$$

Here, λ_q weights state family q , and a family with no variable coordinates contributes zero loss. The corruption and sampling operators assign zero rates to program-fixed coordinates and project each finite-step state back onto them as an implementation safeguard. This is the precise syntactic sense in which the program constrains a state trajectory: fixed coordinates cannot be resampled. Chemical consistency with the requested assembly is established separately by exact forward replay. Under an unrestricted coordinatewise model class, every population minimizer of the primary masked denoising term recovers the corresponding clean-coordinate posterior marginals. This result does not characterize the optimizer of the auxiliary-loss or PCGrad procedure (Propositions E.5 and E.6).

The denoiser is a graph Transformer over the complete sparse state. Each layer applies graph self-attention with relation biases for parent, child and closure edges, then cross-attends to memory tokens representing program identity, program depth, precursor role and reaction-core position. Cross-attention is repeated in every layer so that reaction semantics can influence both local state updates and long-range coordination among precursor-derived regions. Three softly program-routed residual adapters provide family-specific capacity while retaining a shared whole-graph backbone.

For each reaction family f , we compute a task loss

$$\mathcal{L}_f = \mathcal{L}_{\text{FORGE},f} + \lambda_{\text{role}} \mathcal{L}_{\text{role}} + \lambda_{\text{core}} \mathcal{L}_{\text{core}}. \quad (28)$$

Present semantic states receive equal mass within each auxiliary term. Training gives every reaction family equal total loss mass. The gradients of the family task losses, not the individual loss heads, are combined by deterministic PCGrad. Families are ordered by their integer program state. Writing $g_f = \nabla_\theta \mathcal{L}_f$, the algorithm initializes $\tilde{g}_f \leftarrow g_f$ and, for every other family h in that fixed order, applies

$$\tilde{g}_f \leftarrow \tilde{g}_f - \mathbf{1}\{\langle \tilde{g}_f, g_h \rangle < 0\} \frac{\langle \tilde{g}_f, g_h \rangle}{\max(\|g_h\|_2^2, 10^{-12})} g_h \quad (29)$$

on parameter coordinates available to both tasks. Zero gradients remain zero; gradients are not norm-normalized; and the optimizer receives the arithmetic mean of the projected family gradients. This is an algorithmic definition of the update and is not the gradient of the displayed scalar sum in general. Table 7 prespecifies separate interventions for layerwise cross-attention, each auxiliary loss, routed adapters and gradient conflict control.

At inference, FORGE samples X_0 from the factorized program-compatible source, assigns zero update rates to fixed coordinates, and applies L_{samp} capped independent-coordinate Euler steps. At each step the implementation samples one clean category per coordinate from the denoiser, forms the minimum R^* rate for that sampled endpoint, multiplies it by $\Delta t = 1/L_{\text{samp}}$, and rescales the total off-diagonal mass to at most 0.999 before drawing the next coordinate value. This is a finite-step Monte Carlo approximation to the local CTMC, not exact CTMC simulation. Appendix E states the ideal marginalized rate and the numerical kernel. FORGE decodes the terminal state X_1 into a molecular graph G and retains it only if it is connected, valence-valid, and passes molecular sanitization.

B.6 Exact assembly verification

To ensure that each retained graph satisfies the specified reaction program, FORGE decomposes it into role-assigned precursors and applies the corresponding forward transform.

Definition B.3 (Mathematical and verifier-relative L1). For chemistry c , the reverse search $\mathcal{D}_c$ maps G to candidate role-assigned precursor tuples and atom-mapped traces. The complete mathematical preimage $\mathcal{E}_P^*(G)$ contains every admissible input whose registered forward operator can produce G . Define

$$\text{L1}^*(G, P) = \mathbf{1}[\mathcal{E}_P^*(G) \neq \emptyset]. \quad (30)$$

The bounded implementation searches $\mathcal{D}_P(G) \subseteq \mathcal{D}_c(G)$ and returns

$$\widehat{\mathcal{E}}_P(G) = \{(\mathbf{C}, \tau) \in \mathcal{D}_P(G) : \exists H \in \mathcal{F}_P(\mathbf{C}) \text{ such that } H \cong G\}. \quad (31)$$

Its verified predicate is

$$\widehat{\text{L1}}(G, P) = \mathbf{1}[\widehat{\mathcal{E}}_P(G) \neq \emptyset]. \quad (32)$$

548

549 Every returned trace passes exact replay, hence $\widehat{\text{L1}}(G, P) \leq \text{L1}^*(G, P)$. Reverse-search complete-
 550 ness has not been proved, so the reported metric is called *verified exact-L1 yield*. Post-hoc filtering
 551 changes which completed samples are retained but cannot change the proposal distribution’s exact-L1
 552 acceptance probability. Under independent sampling, this probability determines the complete distri-
 553 bution of accepted counts under a fixed attempt budget and the expected cost of obtaining a fixed
 554 number of accepted products (Proposition E.10). FORGE reports decomposition coverage separately
 555 from forward-replay precision because identifying a candidate disconnection does not guarantee
 556 exact reconstruction of G . Candidate decomposition coverage is the fraction of valid graphs for
 557 which $\mathcal{D}_c(G) \neq \emptyset$. Candidate replay precision is

$$\text{ReplayPrecision} = \frac{\sum_G \sum_{(\mathbf{C}, \tau) \in \mathcal{D}_P(G)} \mathbf{1}\{\exists H \in \mathcal{F}_P(\mathbf{C}) : H \cong G\}}{\sum_G |\mathcal{D}_P(G)|}, \quad (33)$$

558 with N/E when the denominator is zero. Replay of tuples already in $\widehat{\mathcal{E}}_P(G)$ is a 100% verifier
 559 invariant rather than an empirical precision result. We separately define

$$\text{CandidateAmbig}(G, c) = \mathbf{1}\{|\mathcal{D}_c(G)| > 1\}, \quad \text{ExactAmbig}(G, P) = \mathbf{1}\{|\widehat{\mathcal{E}}_P(G)| > 1\}. \quad (34)$$

560 A distinct exact-L1 product is a deduplicated generated constitution; an unambiguous exact decompo-
 561 sition instead has $|\widehat{\mathcal{E}}_P(G)| = 1$. We do not use “unique L1” for both.

562 B.7 Recursive synthesis-program construction

563 The final assembly requires access to every recovered precursor. Exact L1 verification identifies
 564 these precursors, but it does not establish whether they are available or how to prepare them. For
 565 laboratory planning, an exact final assembly is useful only when every required precursor is either
 566 available or connected to available starting materials through a supported preparation route. Let
 567 $T = (V_M, V_R, E)$ denote a bipartite synthesis tree rooted at G , where each product molecule points
 568 to its preparation reaction and each reaction points to its required reactants. FORGE constructs
 569 these routes recursively and rejects the lipid when any branch remains unresolved. A molecule node
 570 terminates when a current, dated supplier record lists an exact constitutional match as available;
 571 otherwise, FORGE must expand that node through a supported, forward-verified reaction. FORGE
 572 uses a bounded level-two (L2) planner to propose each expansion. For each proposal, FORGE records
 573 the reaction, reactants, product, and supporting evidence, and accepts the step only when the forward
 574 transform reconstructs the target precursor exactly. Each accepted reaction adds its reactants as new
 575 molecule nodes, and FORGE applies the same termination-or-expansion rule recursively. The current
 576 assessor admits only an unambiguous verified exact-L1 root. Every dossier therefore records the sole
 577 selected root trace $(\mathbf{C}_T, \tau_T) \in \widehat{\mathcal{E}}_P(G)$; an ambiguous exact set produces a typed abstention. For each
 578 upstream reaction node r , let $\text{Admitted}_{\mathfrak{B}}(r) = 1$ when its reaction and substrate scope satisfy the
 579 frozen evidence policy, and let $\text{Verify}_{\mathfrak{B}}(r) = 1$ when $\text{target}(r) \in \mathcal{F}_r(\text{reactants}(r))$. For each leaf
 580 molecule ℓ , let $\text{Available}_{\mathfrak{B}}(\ell) = 1$ when it satisfies the terminal supplier-evidence rule. Let $V_R^{\text{up}}(T)$
 581 denote the upstream preparation-reaction nodes and $L(T)$ the leaf molecules. $\text{WellFormed}(G, T, P)$
 582 requires that T is a connected, acyclic bipartite tree rooted at G ; each internal molecule has exactly
 583 one recorded producing reaction; each reaction’s incoming molecule children equal its recorded
 584 reactants; and T has no disconnected, cyclic or extraneous node. Its distinguished final-assembly

Algorithm 1 FORGE generation and fail-closed synthesis-program construction.

Require: trained coordinate predictors, reaction program P , Euler-step count L_{samp} , attempt budget B , recorded seed s , and frozen route contract $\mathfrak{B}_{\text{route}}$
Ensure: attempt-indexed dossier sequence $\mathcal{D}_{\text{out}}$, exact-L1 ledger $\mathcal{L}_{\text{out}}$, and typed failure ledger $\mathcal{H}$

- 1: initialize the random-number streams from s ; $\mathcal{D}_{\text{out}} \leftarrow []$; $\mathcal{L}_{\text{out}} \leftarrow []$; $\mathcal{H} \leftarrow []$
- 2: **for** $b = 1, \dots, B$ **do**
- 3: sample $X_0 \sim q_0^P$ and set $X \leftarrow \Pi_P(X_0)$
- 4: **for** $k = 0, \dots, L_{\text{samp}} - 1$ **do**
- 5: $t_k \leftarrow k/L_{\text{samp}}$; $\Delta t_k \leftarrow 1/L_{\text{samp}}$
- 6: $X \leftarrow \Pi_P(\text{CAPPEDCOORDINATEEULER}(X, t_k, \Delta t_k, P))$
- 7: **end for**
- 8: $G \leftarrow \text{DECODE}(X)$
- 9: **if** $G = \perp_{\text{dec}}$, is valence-invalid, or fails molecular sanitization **then**
- 10: $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{INVALIDGRAPH})]$
- 11: **continue**
- 12: **end if**
- 13: $\hat{\mathcal{E}}_P(G) \leftarrow \{(\mathbf{C}, \tau) \in \mathcal{D}_P(G) : \exists H \in \mathcal{F}_P(\mathbf{C}), H \cong G\}$
- 14: **if** $\hat{\mathcal{E}}_P(G) = \emptyset$ **then**
- 15: $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{L1FAILURE})]$
- 16: **continue**
- 17: **end if**
- 18: $\mathcal{L}_{\text{out}} \leftarrow \mathcal{L}_{\text{out}} \parallel [(b, G, \hat{\mathcal{E}}_P(G))]$
- 19: **if** $|\hat{\mathcal{E}}_P(G)| \neq 1$ **then**
- 20: $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{AMBIGUOUSL1})]$
- 21: **continue**
- 22: **end if**
- 23: select the sole $(\mathbf{C}_T, \tau_T) \in \hat{\mathcal{E}}_P(G)$
- 24: $Y \leftarrow \text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T)$
- 25: **if** Y is a tree T and $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$ **then**
- 26: $\mathcal{D}_{\text{out}} \leftarrow \mathcal{D}_{\text{out}} \parallel [(b, \text{DOSSIER}(G, P, \mathbf{C}_T, \tau_T, T))]$
- 27: **else**
- 28: $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, Y)]$ $\triangleright$ preserve the typed outcome
- 29: **end if**
- 30: **end for**
- 31: **return** $\mathcal{D}_{\text{out}}, \mathcal{L}_{\text{out}}, \mathcal{H}$

585 record must have product G and precursor children exactly $\mathbf{C}_T$ under trace τ_T .

$$\begin{aligned} \text{Complete}_{\mathfrak{B}}(G, T, P) &= \text{WellFormed}(G, T, P) \mathbf{1}[(\mathbf{C}_T, \tau_T) \in \hat{\mathcal{E}}_P(G)] \\ &\quad \times \prod_{r \in V_R^{\text{up}}(T)} \text{Admitted}_{\mathfrak{B}}(r) \text{Verify}_{\mathfrak{B}}(r) \prod_{\ell \in L(T)} \text{Available}_{\mathfrak{B}}(\ell). \end{aligned} \quad (35)$$

586 Empty products equal one. Let $\mathfrak{B}_{\text{route}}$ fix the route-depth and query budgets, registered transforms,
587 evidence-admission policy, and dated supplier snapshot used by $\text{Assess}_{\mathfrak{B}}$. Under finite search
588 bounds and terminating evidence queries, the assessor terminates and returns a dossier only when
589 $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$ (Lemma E.11 and Invariant E.12). An abstention means that FORGE
590 did not complete a dossier under $\mathfrak{B}_{\text{route}}$; it does not imply that no synthesis route exists outside the
591 frozen search space or evidence snapshot. The guarantee is computational and does not establish
592 experimental synthesis success.

593 Algorithm 1 summarizes the complete inference procedure. Every iteration consumes one generation
594 attempt, and FORGE retains the outcome of every failed attempt in a typed ledger.

595 For every accepted pair (G, T) , FORGE records a computational synthesis dossier containing all
596 molecules and reactions in T , the evidence supporting each reaction, and the dated supplier record for
597 every terminal material. We classify each generated lipid into one of four support tiers according to
598 which elements of its assembly lie outside the fixed reference space: **E0 (enumerated)**: the generated
599 lipid matches a product that was already constructed computationally from known components

600 using a supported reaction. **E1 (known components):** the generated lipid was not included in the
601 bounded E0 enumeration, but a supported reaction assembles it entirely from known components.
602 **E2 (generated precursor):** a supported reaction assembles the lipid, but at least one precursor is
603 newly generated and requires a complete L2 route to available starting materials. **E3 (new assembly**
604 **family):** the lipid requires a final assembly reaction outside the qualified reaction registry.

C Extended experimental protocol and results

This appendix gives the full reporting contract, including every matched arm, ablation and diagnostic summarized in Section 3 of the body.

C.1 Experimental settings

Our evaluation spans the three assembly programs defined above, with Ugi as the deeply supervised case, repeated aza-Michael addition as a second family, and repeated reductive amination as a lighter stress test. The evaluation tests four capabilities: learning multiple assembly programs within one model, using reaction semantics during generation rather than after it, recovering verified decompositions with components absent from the training catalogue, and tracing generated precursors to available starting materials. The final matched comparison uses three shared-model arms: conditioning across all three programs, training without program information, and training with cyclically incorrect program labels. All arms use the same architecture and optimization budget, and we evaluate the prespecified final checkpoint from each of three independent seeds. Each matched comparison uses the same number of generation attempts per program. Our primary metric is verified exact-L1 yield per generation attempt: the number of generated graphs that pass molecular validity, program-aware decomposition, and exact forward replay divided by the total number of sampling attempts. Secondary metrics separate graph quality (validity and connectedness), reaction recovery (verified coverage, the replay invariant, abstention, and exact ambiguity), and exploration (distinct-product fraction, whole-product and component novelty, molecular diversity, and effective component count). We report exact counts, means across seeds, sample standard deviations, individual seed values, and the minimum and maximum of the three paired seed contrasts. These are descriptive across-seed summaries, not confidence intervals. Generated molecules are not treated as independent training replicates. Conditional binomial variation across a fixed model’s 3,072 attempts is distinct from variation induced by retraining. Training uses only the component-disjoint training folds. Calibration data are diagnostic, and held-out products neither enter training nor select checkpoints. The three shared arms assign equal total sampling mass to each program while preserving the frozen within-program source weights. The frozen implementation evaluated numerical promotion gates, but three seeds do not support a population-level non-inferiority claim. We therefore report only whether the observed runs met each numeric rule. Zero program-fixed-coordinate violations are an implementation check required by Proposition E.5, not learned behavior.

Two additional comparisons isolate the role of reaction semantics. First, we compare conditioned generation with null generation followed by exact post-hoc reaction filtering under the same attempt budget; the cyclic-label arm tests whether chemical meaning matters beyond the presence of a categorical input. Second, we compare FORGE with a train-only finite-component assembler that receives the exact training components, supported forward chemistry, source-weighted role marginals, and the same number of attempts. We report reaction validity and exploration separately because the catalogue can achieve high exact-L1 yield while having zero input-component novelty by construction. These comparisons make no L2 planner, biological-oracle, or candidate-selection calls.

For the Ugi program, we additionally define a common external-baseline protocol with one reaction-space comparator, RGFN, and two generic whole-molecule comparators, unconditional DeFoG and GenMol/SAFE [Kozlarski et al., 2024, Qin et al., 2025, Lee et al., 2025]. The Synthesis-DAG method of Ou et al. [2024], SynFlowNet and SynCoGen remain methodological context rather than empirical rows [Cretu et al., 2025, Rekesh et al., 2025]. The released Synthesis-DAG repository does not contain the paper-specific ionizable-lipid implementation, the released SynFlowNet action space cannot represent the qualified three-reactant Ugi transform, and the released SynCoGen repository has no usable code license. We do not replace missing functionality with an imitation, alter Ugi into sequential pseudo-reactions, or execute unlicensed code. An external row enters the numerical comparison only after the method has been executed on the same training split, generation-attempt budget, seed set, and common molecular and exact-L1 evaluation pipeline used for FORGE. Published values obtained on the method’s original dataset are not mixed with results from this common evaluation. Methods that do not support the additional assembly programs are compared on Ugi only rather than assigned artificial aza-Michael or reductive-amination results. The matched internal suite separately includes unconditioned generation plus post-hoc filtering, an oracle finite-catalogue assembler, a learned inventory selector, two learned component-factorized generators, and targeted interventions on the Transformer. Table 7 reports the completed three-seed Transformer

mechanism and component-factorization study. Each intervention uses the same folds, generation budget and evaluation code as the full Transformer.

C.2 Metric, identity and uncertainty definitions

Let $Y_{s,a,b}^{\text{L1}} \in \{0, 1\}$ indicate verified exact L1 for training seed s , arm a and attempt b , and let $B = 3,072$. We report the exact count $C_{s,a}^{\text{L1}} = \sum_{b=1}^B Y_{s,a,b}^{\text{L1}}$ and

$$\hat{p}_{s,a}^{\text{L1}} = C_{s,a}^{\text{L1}}/B. \quad (36)$$

Validity and connected yield replace Y^{L1} with their attempt-level indicators. Verified decomposition coverage is the number of valid graphs with $\hat{\mathcal{E}}_P(G) \neq \emptyset$ divided by valid graphs; the implementation does not report candidate-search coverage before replay. Exact ambiguity is the fraction of valid graphs with $|\hat{\mathcal{E}}_P(G)| > 1$. Replay of members of $\hat{\mathcal{E}}_P(G)$ is an invariant, while candidate replay precision is defined in Appendix B and is N/E if no candidate tuple is returned. Explicitly, with attempt outcomes $G_{s,a,b}$ and the convention that a failed decode makes every graph predicate zero,

$$\hat{p}_{s,a}^{\text{valid}} = B^{-1} \sum_b \mathbf{1}\{\text{Valid}(G_{s,a,b}) = 1\}, \quad (37)$$

$$\hat{p}_{s,a}^{\text{conn}} = B^{-1} \sum_b \mathbf{1}\{\text{Valid}(G_{s,a,b}) = 1, \text{Connected}(G_{s,a,b}) = 1\}, \quad (38)$$

$$\text{CandidateCoverage} = \frac{\sum_b \mathbf{1}\{\text{Valid}(G_b) = 1, \mathcal{D}_c(G_b) \neq \emptyset\}}{\sum_b \mathbf{1}\{\text{Valid}(G_b) = 1\}}, \quad (39)$$

$$\text{ExactAmbiguity} = \frac{\sum_b \mathbf{1}\{\text{Valid}(G_b) = 1, |\hat{\mathcal{E}}_P(G_b)| > 1\}}{\sum_b \mathbf{1}\{\text{Valid}(G_b) = 1\}}. \quad (40)$$

Each ratio is N/E when its denominator is zero.

Distinct-product yield is the number of distinct canonical product constitutions satisfying verified exact L1 divided by B . The reported verifier-recovered component-novel yield additionally requires $|\hat{\mathcal{E}}_P(G)| = 1$ and at least one returned role-assigned precursor constitution absent from the fixed training catalogue. It is not StrongEscape. Held-component recovery counts verified exact-L1 attempts whose returned tuple contains a designated held constitution and divides by B . Writing $\text{can}(G)$ for constitutional canonicalization, $\mathcal{C}_{\text{train}}$ for the role-indexed training catalogue and $\mathcal{H}_{\text{held}}$ for the designated held constitutions, these yields are

$$\text{DistinctL1Yield} = B^{-1} \left| \left\{ \text{can}(G_b) : \widehat{\text{L1}}(G_b, P) = 1 \right\} \right|, \quad (41)$$

$$\begin{aligned} \text{ComponentNovelYield} = B^{-1} \sum_b \mathbf{1} \Big[& \widehat{\text{L1}}(G_b, P) = 1, |\hat{\mathcal{E}}_P(G_b)| = 1, \\ & \exists(r, C) \in \mathbf{C}_b : C \notin \mathcal{C}_{\text{train},r} \Big], \end{aligned} \quad (42)$$

$$\text{HeldComponentYield} = B^{-1} \sum_b \mathbf{1} \Big[\widehat{\text{L1}}(G_b, P) = 1, \mathbf{C}_b \cap \mathcal{H}_{\text{held}} \neq \emptyset \Big]. \quad (43)$$

For component frequencies w_i , the effective count is

$$N_{\text{eff}} = \exp \left(- \sum_i w_i \log w_i \right). \quad (44)$$

Component duplicates are retained when estimating w_i . Molecular diversity is the mean pairwise ECFP4 Tanimoto distance among distinct valid canonical product constitutions, so molecular duplicates are removed for that metric. For distinct products $g_1, \dots, g_m$,

$$\text{Diversity} = \frac{2}{m(m-1)} \sum_{1 \leq i < j \leq m} [1 - \text{Tanimoto}(\text{ECFP4}(g_i), \text{ECFP4}(g_j))], \quad (45)$$

with N/E for $m < 2$. Every other undefined conditional denominator is likewise reported N/E.

Table 3: **Matched shared-model production comparison across reaction programs.** Point estimates report the mean across three independent seeds; the exact-L1 column additionally reports the sample standard deviation. Paired contrasts are summarized by their three seed values, mean and observed range; individual seed values appear in the supplement. Validity and connectedness use all generation attempts as the denominator. Decomposition coverage, replay precision, abstention and ambiguity retain their prespecified conditional denominators. Novelty is conditioned on valid products or unique returned decompositions, as appropriate. Fraction pairs and exact-L1 yield are percentages; diversity is mean pairwise ECFP4 distance and N_{eff} is the effective generated-component count. Results are never pooled across reaction programs.

	Ugi			Aza-Michael			Reductive amination		
	FORGE	Shared null	Cyclic	FORGE	Shared null	Cyclic	FORGE	Shared null	Cyclic
Valid./Conn.	96.5/96.5	35.9/35.9	93.9/93.9	72.2/72.2	27.4/27.4	71.5/71.5	60.4/60.4	41.9/41.9	59.1/59.1
Decomp./Replay	99.9/100.0	97.0/100.0	99.8/100.0	100.0/100.0	69.5/100.0	100.0/100.0	86.6/100.0	88.4/100.0	85.7/100.0
Abst./Ambig.	0.1/0.0	3.0/0.9	0.2/0.0	0.0/61.0	30.5/29.3	0.0/62.3	13.4/74.7	11.6/76.6	14.3/73.8
Exact-L1/attempt	96.4 $\pm$ 2.1	34.8 $\pm$ 7.1	93.8 $\pm$ 4.2	72.2 $\pm$ 1.6	19.2 $\pm$ 4.1	71.5 $\pm$ 2.2	52.3 $\pm$ 3.4	37.0 $\pm$ 2.2	50.7 $\pm$ 4.9
Whole/Comp. novelty	99.5/96.7	99.9/99.0	99.4/96.9	89.8/85.3	99.6/99.7	90.2/86.3	67.4/97.5	91.5/99.8	66.0/97.0
Diversity/ N_{eff}	0.643/419.9	0.680/439.5	0.646/427.9	0.593/118.0	0.707/413.5	0.603/119.4	0.776/173.0	0.761/193.3	0.773/157.8

Identity is constitutional and stereo-free. Graph and product identity use connected canonical RDKit SMILES with atom maps and stereochemical tags removed; formal charges and bond orders are retained; salts or disconnected products are invalid; and no tautomer normalization is applied. Precursor identity uses the same canonicalization. Precursor-tuple equality is role- and program-step-aware, with repeated inputs compared in their registered step order. Atom mappings and the trace τ are evidence for semantic alignment, not part of constitutional identity.

For a paired contrast, $D_s = \hat{p}_{s,\text{FORGE}} - \hat{p}_{s,\text{control}}$. We report (D_1, D_2, D_3) , $\bar{D}$, and $[\min_s D_s, \max_s D_s]$. These summarize the three observed training seeds and are not 95% confidence intervals. Conditional on one fixed trained model, $C_{s,a}^{\text{L1}}$ has the finite-attempt binomial variation in Proposition E.10; that uncertainty does not substitute for independent retraining. Unless explicitly named primary, architecture, coordinate, guidance, potency and catalogue contrasts are exploratory and no multiplicity-adjusted inferential claim is made.

C.3 Shared reaction-program learning across three assembly chemistries

Across 3,072 held-out generation attempts per program in each of three independent training seeds, the final conditioned model achieved verified exact-L1 yields of $96.4 \pm 2.1\%$ for Ugi, $72.2 \pm 1.6\%$ for repeated aza-Michael addition and $52.3 \pm 3.4\%$ for repeated reductive amination (mean $\pm$ sample standard deviation; Table 1 and Table 3). Every verifier-returned trace passed forward replay, as required by the verifier invariant. However, mean exact- decomposition ambiguity was 61.0% for repeated aza-Michael addition and 74.7% for repeated reductive amination, and the latter program showed substantial between-seed variation. Table 3 reports molecular validity, connectedness, decomposition coverage, forward-replay precision, abstention, ambiguity, novelty and diversity separately for each program.

The final conditioned model recorded zero fixed-coordinate violations and zero support overflows across 27,648 held-out attempts (three seeds, three programs and 3,072 attempts per program). Together, these results support the conclusion that one conditional flow can learn the three specified reaction-program coordinates, while performance remains reaction-family dependent.

Table 4: **Decoder and source-marginal ablations of the final model.** Ugi held-out attempts for the 9,143-step shared model, which is the model reported elsewhere in this paper. The two ablations are independent: the decoder rows share one set of trained weights and differ only in terminal decoding, while the source rows share one decoder and differ only in the marginal the corruption starts from. Decoding under the reaction-core saturation policy holds the aldehyde-derived α -carbon at one hydrogen and the isocyanide-derived amide nitrogen at one hydrogen, which the qualified transform fixes; it changes which graph is produced, never how it is adjudicated. Values are percentages from 3,072 seed-0 held-out attempts per row; they are descriptive and do not measure across-training-seed variation. Exact-L1 coverage is conditioned on valid products.

Source marginal	Terminal decoder	Valid	Exact-L1 coverage	Exact-L1
Program-role	Core saturation	95.21%	99.90%	95.12%
Program-role	Valence topology only	95.31%	78.35%	74.67%
Global	Core saturation	86.95%	99.63%	86.62%

Table 5: **Exact verified-L1 counts for the final conditioned model.** Each row reports the numerator, fixed attempt denominator and resulting percentage for one independently trained seed. These attempt-level counts quantify conditional Monte Carlo variation; the three seeds, rather than the individual attempts, are the units for retraining variation. Seed labels 0, 1 and 2 correspond to RNG seeds 20260825, 20260826 and 20260827, respectively.

Program	Seed	Verified L1 count	Attempts	Yield (%)
Ugi	0	2,922	3,072	95.12
Ugi	1	3,037	3,072	98.86
Ugi	2	2,924	3,072	95.18
Aza-Michael	0	2,176	3,072	70.83
Aza-Michael	1	2,203	3,072	71.71
Aza-Michael	2	2,274	3,072	74.02
Reductive amination	0	1,561	3,072	50.81
Reductive amination	1	1,727	3,072	56.22
Reductive amination	2	1,535	3,072	49.97

710 C.4 Reaction semantics alter generation rather than merely filtering its output

711 Under matched attempt budgets, conditioned generation produced $96.4 \pm 2.1\%$, $72.2 \pm 1.6\%$ and
 712 $52.3 \pm 3.4\%$ verified exact-L1 yield for Ugi, repeated aza-Michael and repeated reductive amination,
 713 compared with $34.8 \pm 7.1\%$, $19.2 \pm 4.1\%$ and $37.0 \pm 2.2\%$ for matched null generation followed
 714 by exact verification (Table 3). The paired conditioned-minus-null differences were 61.6 percentage
 715 points for Ugi (three-seed range 57.4 to 67.7), 53.0 for aza-Michael (50.2 to 56.4), and 15.4 for
 716 reductive amination (14.5 to 16.7).

717 The cyclic control retained precursor roles, reaction-core coordinates and program depth but assigned
 718 the wrong reaction identity. Relative to correct conditioning, this intervention changed verified
 719 exact-L1 yield by 2.6 percentage points for Ugi (three-seed range 0.7 to 5.5), 0.6 for aza-Michael
 720 (-2.4 to 3.8), and 1.6 for reductive amination (-0.1 to 2.5). Every returned exact trace passed replay
 721 by verifier construction.

722 Because the null and cyclic arms are independently retrained under the same effective study contract,
 723 this comparison measures the effect of program information on learned generation rather than relying
 724 on an intervention performed on an older checkpoint family. The complete coordinate set has a large
 725 contrast against the null arm, but the small cyclic-label contrasts do not identify a strong independent
 726 contribution from the correct reaction-identity token.

727 C.5 Whole-graph generation recovers precursor identities absent from the training catalogue

728 Table 2 defines the common Ugi benchmark used to compare FORGE with external synthesis-aware
 729 generators, matched internal controls, and the finite-catalogue oracle. Every yield is normalized by
 730 the number of generation attempts, including attempts that fail before producing a valid molecule.
 731 All included methods were run under the common split, attempt budget, seed set and verifier.

Table 6: **Method-blind structural-realism diagnostic.** Every entry is mean $\pm$ sample standard deviation across three training seeds and retains all 3,072 requested attempts in the denominator. FP and descriptor P/C report manifold precision per 1,000 attempts and coverage of the source-study-held-out R0 reference in percent. C2ST is grouped classifier two-sample area under the receiver operating characteristic curve; 0.5 is indistinguishable. N_{eff} is the Shannon effective molecule count. N/E denotes a metric that cannot be estimated. This post-hoc structural diagnostic is not a property, activity or synthesis-success score.

Method	Connected/1k	FP P/C	Desc. P/C	C2ST	N_{eff}	Diversity
RGFN	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	N/E	0.0 $\pm$ 0.0	N/E
DeFoG unconditional	346.5 $\pm$ 42.3	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	17.3 $\pm$ 1.4/1.45 $\pm$ 0.06	0.961 $\pm$ 0.007	1064.3 $\pm$ 129.9	0.726 $\pm$ 0.013
GenMol/SAFE	627.5 $\pm$ 6.2	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	N/E	3.2 $\pm$ 0.1	0.873 $\pm$ 0.009
Finite catalogue oracle	1000.0 $\pm$ 0.0	0.2 $\pm$ 0.2/0.05 $\pm$ 0.05	54.6 $\pm$ 4.4/2.02 $\pm$ 0.17	0.971 $\pm$ 0.007	2395.5 $\pm$ 31.7	0.720 $\pm$ 0.000
Learned inventory selector	1000.0 $\pm$ 0.0	0.1 $\pm$ 0.2/0.02 $\pm$ 0.03	56.6 $\pm$ 7.2/1.97 $\pm$ 0.46	0.974 $\pm$ 0.002	2443.0 $\pm$ 55.8	0.718 $\pm$ 0.001
FORGE Transformer	965.0 $\pm$ 21.9	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	29.5 $\pm$ 2.6/2.21 $\pm$ 0.03	0.972 $\pm$ 0.004	2489.2 $\pm$ 66.8	0.642 $\pm$ 0.022

The learned finite-inventory selector achieved 1000.0 ± 0.0 verified exact-L1 products per 1,000 attempts, but its finite action space imposed zero verifier-recovered component-novel yield. FORGE produced 963.9 ± 21.4 verified exact-L1 products and 862.6 ± 19.2 distinct verified exact-L1 products per 1,000 attempts. Among whole-molecule external generators, unconditional DeFoG produced 280.6 ± 26.6 verified exact-L1 products per 1,000 attempts and recovered held components at 15.5 ± 2.4 per 1,000 attempts, compared with 1.4 ± 1.7 for FORGE. GenMol/SAFE produced 627.5 ± 6.2 valid connected outputs but no exact Ugi-L1 product, while RGFN emitted no admissible molecular output under the strict native contract. These outcomes expose separate tradeoffs between reaction consistency, verifier-recovered component novelty and recovery of specifically reserved components. They do not establish product-level strong catalogue escape.

No evaluated method reproduced the full held-out observed-lipid distribution: where estimable, grouped C2ST AUC remained between 0.961 and 0.974, and fingerprint-manifold precision was zero or near zero (Table 6). GenMol/SAFE’s high pairwise diversity was paired with an effective count of only 3.2 ± 0.1 molecules, confirming severe frequency collapse. FORGE retained 2489.2 ± 66.8 effective molecules, but only 29.5 ± 2.6 attempts per 1,000 entered the descriptor manifold. We therefore interpret this analysis as a multi-axis structural diagnostic rather than a single measure of whether a generated lipid is reasonable.

C.6 Joint whole-lipid denoising has reaction-dependent advantages over role isolation

The matched-study full Transformer produced 856.9 Ugi and 767.7 repeated aza-Michael exact-L1 products per 1,000 attempts, compared with 56.9 and 396.4 for the parameter-matched role-isolated FACT model (Table 7). The full Transformer and FACT-matched model each had 5.33 million trainable parameters; FACT-generous had 29.40 million. The higher-capacity FACT control improved to 455.0 for Ugi and 685.7 for repeated aza-Michael addition. Its three-seed means remained below the full Transformer for both programs, although the aza-Michael values were essentially tied in one seed. The direction reversed for repeated reductive amination: the full Transformer produced 318.8 exact-L1 products per 1,000 attempts, whereas FACT-matched and FACT-generous produced 520.3 and 558.1. Thus, joint whole-lipid denoising improved exact assembly for Ugi and repeated aza-Michael addition but was not uniformly superior across reaction programs. Because the prespecified conditional-dependence diagnostic was negative, this architecture contrast does not establish detectable cross-role dependence in the present data.

Removing layerwise program cross-attention reduced mean exact-L1 yield by 61.6 products per 1,000 attempts for Ugi and 50.7 for repeated reductive amination, while increasing repeated aza-Michael yield by 8.8. Removing the role or core objective and removing routed adapters also produced program-dependent changes rather than a uniform loss. The arm without gradient-conflict control slightly exceeded the full Transformer mean in all three programs. We therefore do not claim that every introduced module independently improves exact-L1 yield. These one-at-a-time interventions identify which effects replicate across chemistries; they are not pooled into a composite architecture score.

At the matched sampling budget, FORGE produced 839.2 verifier-recovered component-novel Ugi products per 1,000 attempts, averaged across the three independent seeds. The corresponding

Table 7: **Transformer architecture and mechanism ablations.** The five Transformer intervention rows change only the named mechanism from the matched-study full model; the two FACT rows instead generate precursor roles independently at matched or increased capacity. All arms use identical data folds, corruption, context schedule, checkpoint rule, core-saturation terminal decoder and sampling budget. The reference full Transformer is the matched mechanism-study control, not the independently trained production model reported in Tables 3 and 8. Entries report the mean $\pm$ sample standard deviation across three independent training seeds. Held-out loss is the mean denoising loss at $t = 0.5$ across the three component-disjoint program partitions. Cross-role fidelity is the Frobenius distance to the held-out residualized descriptor matrix, for which lower values indicate closer agreement. Exact-L1 yields use all 3,072 attempts per program and seed as the denominator.

Arm	Held-out loss	Ugi L1/1k	BL L1/1k	LX L1/1k	Held-comp. L1/1k	Cross-role fidelity	Diversity
Input-only program	8.48 $\pm$ 0.72	795.2 $\pm$ 112.4	776.5 $\pm$ 33.2	268.1 $\pm$ 56.9	0.00 $\pm$ 0.00	1.83 $\pm$ 0.15	0.597 $\pm$ 0.009
No role loss	7.87 $\pm$ 0.12	846.2 $\pm$ 98.2	769.6 $\pm$ 45.6	289.7 $\pm$ 148.1	0.22 $\pm$ 0.19	1.92 $\pm$ 0.07	0.587 $\pm$ 0.011
No core loss	7.85 $\pm$ 0.11	833.4 $\pm$ 101.7	776.0 $\pm$ 37.2	310.1 $\pm$ 130.1	0.22 $\pm$ 0.19	1.82 $\pm$ 0.15	0.584 $\pm$ 0.007
No routed adapters	7.81 $\pm$ 0.32	791.0 $\pm$ 164.9	791.0 $\pm$ 49.5	329.9 $\pm$ 60.9	0.11 $\pm$ 0.19	1.77 $\pm$ 0.07	0.587 $\pm$ 0.001
No gradient conflict control	7.98 $\pm$ 0.20	866.4 $\pm$ 84.1	774.1 $\pm$ 44.2	324.1 $\pm$ 157.4	0.11 $\pm$ 0.19	1.92 $\pm$ 0.12	0.584 $\pm$ 0.008
FACT-matched	7.09 $\pm$ 0.24	56.9 $\pm$ 24.4	396.4 $\pm$ 63.5	520.3 $\pm$ 62.2	0.11 $\pm$ 0.19	1.31 $\pm$ 0.22	0.555 $\pm$ 0.014
FACT-generous	8.24 $\pm$ 0.31	455.0 $\pm$ 191.9	685.7 $\pm$ 107.4	558.1 $\pm$ 63.6	1.95 $\pm$ 0.86	1.12 $\pm$ 0.04	0.589 $\pm$ 0.005
FORGE	7.88 $\pm$ 0.22	856.9 $\pm$ 86.7	767.7 $\pm$ 35.9	318.8 $\pm$ 158.3	0.11 $\pm$ 0.19	1.90 $\pm$ 0.13	0.585 $\pm$ 0.007

values were 159.3 for aza-Michael addition and 58.7 for reductive amination. The finite-component assembler produced zero by construction because every sampled product uses only components from the training catalogue (Table 8). The catalogue achieved exact-L1 yields of 1000.0, 723.2 and 1000.0 products per 1,000 attempts, compared with 963.9, 721.9 and 523.3 for FORGE. The corresponding raw validities were 1000.0, 723.2 and 1000.0 for the catalogue and 965.0, 722.2 and 604.3 for FORGE. We report these quantities separately because reaction validity and verifier-relative component novelty measure different capabilities. Direct membership in the complete catalogue-reachable product set was not computed, so these results are not reported as strong catalogue escape.

In a distinct-product Ugi role audit, 19,520 of 26,235 products (74.4%) contained at least one verifier-returned component constitution absent from the training registry. Across all Ugi products admitted to the novelty audit, 21,960 of 26,235 distinct products (83.7%) were absent from the full 112,386-product constitutional reference corpus. FORGE effective component counts were 419.9, 118.0 and 173.0, compared with 86.4, 23.4 and 16.9 for the finite catalogue. The Ugi catalogue contained 185 amines, 75 aldehydes and 35 isocyanides and covered 100% of training products but 0% of the calibration and component-disjoint held-out products. Its Cartesian tuple ceilings were 485,625 for Ugi, 774 for aza-Michael and 140 for reductive amination; these count possible component-program inputs rather than unique forward products. The exact finite catalogue has high transform consistency, whereas whole-graph generation can produce verified exact-L1 products with verifier-recovered precursor identities absent from that inventory.

Table 8: **Whole-graph generation versus a finite-component assembler.** Verified exact-L1, distinct exact-L1 and verifier-recovered component-novel yields are reported per 1,000 generation attempts. A component-novel product has exactly one verifier-returned exact-L1 decomposition and contains at least one component constitution absent from the training catalogue. This verifier-relative metric is not strong catalogue escape because complete catalogue-product reachability was not computed. The catalogue has zero input-component novelty by construction, but may achieve higher raw validity or exact-L1 yield. FORGE-minus-catalogue contrasts use the three seeds as descriptive paired units; no three-seed confidence interval is inferred.

	Ugi		Aza-Michael		Reductive amination	
	FORGE	Finite cat.	FORGE	Finite cat.	FORGE	Finite cat.
Valid/1k	965.0	1000.0	722.2	723.2	604.3	1000.0
Verified exact L1/1k	963.9	1000.0	721.9	723.2	523.3	1000.0
Distinct exact L1/1k	862.6	837.6	566.1	147.5	270.6	52.1
Verifier-recovered component-novel/1k	839.2	0.0	159.3	0.0	58.7	0.0
Whole novelty	99.5	16.9	89.8	39.8	67.4	44.8
Component novelty	96.7	0.0	85.3	0.0	97.5	0.0
Diversity/ N_{eff}	0.643/419.9	0.719/86.4	0.593/118.0	0.402/23.4	0.776/173.0	0.514/16.9

Table 9: **Common synthesis-evidence assessment of Ugi generative methods.** The finite-vocabulary field records whether generation is restricted to a fixed component inventory. Numerical evidence fields are populated only after applying the same method-blind union of exact-L1 components, upstream forward-verification, terminal-evidence, and abstention policy to each method’s outputs. All numerical fields are reported per 1,000 generation attempts as mean $\pm$ sample standard deviation across three seeds. A zero downstream count means that no attempt reached that evidence state under the frozen policy. An abstention records missing exact evidence, not a failed synthesis; the frozen exact-source component library is deliberately non-exhaustive.

Method	Finite vocab.	Verified L1/1k	Verified upstream	Terminal evidence	Complete dossier	Abstention
RGFN	yes	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
DeFoG unconditional	no	280.6 $\pm$ 26.6	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	280.6 $\pm$ 26.6
GenMol/SAFE	no	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Finite catalogue oracle	yes	1000.0 $\pm$ 0.0	0.1 $\pm$ 0.2	0.1 $\pm$ 0.2	0.1 $\pm$ 0.2	999.9 $\pm$ 0.2
Learned inventory selector	yes	1000.0 $\pm$ 0.0	0.7 $\pm$ 0.3	0.7 $\pm$ 0.3	0.7 $\pm$ 0.3	999.3 $\pm$ 0.3
Shared-null + post-hoc	no	114.4 $\pm$ 10.9	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	114.4 $\pm$ 10.9
FACT-matched	no	52.0 $\pm$ 15.6	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	52.0 $\pm$ 15.6
FORGE Transformer	no	963.9 $\pm$ 21.4	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	963.9 $\pm$ 21.4

C.7 Native recovery of held components remains weak

The component-disjoint Ugi fold was never used for training, checkpoint selection or inventory construction. Under the common native-generation assessment, the full Transformer recovered only 13 exact-L1 products containing a designated held component in 9,216 attempts, or 1.4 ± 1.7 products per 1,000 attempts averaged across three seeds. This is a negative generalization result. It does not conflict with the broader verifier-relative novelty measurements above: generating a component absent from the training registry is easier than recovering one of the specifically reserved held-component identities. We therefore do not claim strong held-component recovery, role-specific held-component coverage, or zero-shot held-reaction-family generalization.

C.8 Recursive synthesis programs connect generated lipids to evidence-qualified materials

Table 9 reports each method’s finite-vocabulary constraint separately from evidence obtained under the common FORGE assessor. We freeze the union of exact-L1 components from all methods and seeds before route assessment, hide method membership from the evidence pass, and require one complete, unresolved or search-censored disposition for every component before unsealing membership. Emitting a reaction sequence is not counted as a complete dossier unless the final assembly and every upstream step pass exact forward verification and every terminal material closes under the frozen evidence policy. We do not report a separate AiZynthFinder row: a proposal-only route is a hypothesis and cannot change exact evidence closure until every step passes the same forward-verification and terminal-evidence gates [Genheden et al., 2020].

The frozen union contained 11,021 distinct role–constitution components, but the exact-source library closed only 30. Consequently, FORGE produced 963.9 ± 21.4 verified exact-L1 products per 1,000 attempts, yet 0.0 ± 0.0 complete dossiers and 963.9 ± 21.4 abstentions under this library. The learned inventory selector closed 0.7 ± 0.3 dossiers per 1,000 attempts through occasional reuse of covered training components. This is a negative result about current exact evidence coverage, not a comparison of intrinsic synthesizability: components outside the frozen library were not searched and remain unknown.

The final-model route assessment starts from a deterministic, route-blind sample of 256 distinct products drawn by canonical-product hash from 7696 valid, verified exact-L1 products across the three final training seeds. All 256 products retained one exact forward-verified root trace and entered the frozen evidence lookup. 0 products closed under the exact-source and terminal-evidence policy, 256 remained unresolved and 0 were censored. The products contained 379 unique role–constitution component targets: 4 complete, 375 unresolved and 0 censored. No planner or proposal engine was called during scoring.

Table 10: **Route-blind assessment returns auditable outcomes for final Transformer products.** The deterministic hash-ranked sample contains distinct Ugi products with one exact forward-verified L1 trace. Complete denotes closure of every recovered component under the frozen exact-source upstream-reaction and terminal-evidence policy. Unresolved and censored outcomes are abstentions, not evidence of unsynthesizability.

Unit	Population	Complete	Unresolved	Censored
Products	256	0	256	0
Unique components	379	4	375	0

818 **C.9 Guidance diagnostics define the limits of additional optimization**

819 At synthesis-guidance strength 0.25, the matched controller evaluated 64 terminal particles and
 820 retained 56 canonical representatives per arm. It changed only 8 terminal identities and returned 2
 821 unique route-ready products, compared with 3 after post-hoc assessment. Only 5 of 48 four-particle
 822 morphology groups contained mixed route-readiness values, leaving most ancestry decisions uniform.
 823 This negative result does not support promoting mid-trajectory synthesis tilting. The production
 824 method therefore remains complete generation followed by proposal-augmented L1/L2/L3 route
 825 assessment.

826 We evaluate biological guidance only in the bounded HeLa messenger-transfection-potential (mTP)
 827 diagnostic. A group-disjoint preterminal classifier detected weak morphology-level signal, with
 828 ROC AUC 0.623 (95% interval 0.565 to 0.679) and a 1.177-fold observed top-quartile gain. We
 829 then compared a frozen applicability-enriched morphology proposal with a nested potency-enriched
 830 proposal using 3,072 generator calls per arm. Table 11 reports the matched terminal outcome.

Table 11: **Predicted HeLa messenger-transfection potential under matched generation budgets.** The support-enriched arm uses the already promoted applicability proposal; the nested arm reallocates morphology mass using the frozen weak potency signal. Conservative-high counts are computational predictions, not transfection measurements. The comparison uses the same generator-call and oracle budgets.

Arm	Attempts	Verified L1	Eligible before budget	Oracle budget	Unique conservative-high
Support-enriched	3,072	2955	90	38	27
Nested potency proposal	3,072	2962	82	38	23

831 The nested-minus-support difference was -1.30 conservative-high products per 1,000 generator calls
 832 (95% interval -2.28 to 0.98). The promotion gate failed. These computations evaluate predictor
 833 behavior within a declared applicability domain; they do not establish improved biological activity,
 834 delivery or experimental potency.

835 D Extended computational reporting tables

836 The following tables define the complete numerical reporting contract for the common Ugi bench-
 837 mark. Completed matched controls, the learned inventory selector and three external generators
 838 are populated from hash-pinned three-seed results. The method-blind L2/L3 evidence assessment
 839 is reported in Table 9. Compute accounting remains in the generated evidence receipt; unavailable
 840 training-time fields preclude a cross-method compute-matching claim.

Table 12: **Seed-level common Ugi benchmark results.** Rows are expanded once a method has a hash-verified three-seed result vector. The main table reports the corresponding mean $\pm$ sample standard deviation; generated molecules are not treated as independent training replicates.

Method	Seed(s)	Valid/1k	Verified L1/1k	Distinct L1/1k	Comp.-novel/1k	Held-comp. L1/1k	Diversity
RGFN	20260825	0.0	0.0	0.0	0.0	0.0	N/E
RGFN	20260826	0.0	0.0	0.0	0.0	0.0	N/E
RGFN	20260827	0.0	0.0	0.0	0.0	0.0	N/E
DeFoG unconditional	20260825	299.2	251.0	251.0	251.0	13.7	0.711
DeFoG unconditional	20260826	359.7	288.4	288.4	288.4	18.2	0.733
DeFoG unconditional	20260827	380.5	302.4	302.4	302.4	14.6	0.736
GenMol/SAFE	20260825	627.3	0.0	0.0	0.0	0.0	0.883
GenMol/SAFE	20260826	633.8	0.0	0.0	0.0	0.0	0.869
GenMol/SAFE	20260827	621.4	0.0	0.0	0.0	0.0	0.866
Learned inventory selector	20260825	1000.0	1000.0	863.0	0.0	0.0	0.717
Learned inventory selector	20260826	1000.0	1000.0	835.6	0.0	0.0	0.717
Learned inventory selector	20260827	1000.0	1000.0	849.9	0.0	0.0	0.715
Finite catalogue oracle	20260825	1000.0	1000.0	841.5	0.0	0.0	0.719
Finite catalogue oracle	20260826	1000.0	1000.0	843.1	0.0	0.0	0.719
Finite catalogue oracle	20260827	1000.0	1000.0	828.1	0.0	0.0	0.719
Shared-null + post-hoc	20260825	183.9	114.3	109.7	109.7	0.3	0.738
Shared-null + post-hoc	20260826	187.5	103.5	98.3	98.3	0.7	0.763
Shared-null + post-hoc	20260827	212.6	125.3	117.5	117.5	0.3	0.761
FACT-matched	20260825	59.2	49.5	48.8	48.8	0.0	0.676
FACT-matched	20260826	81.7	68.7	66.1	66.1	0.0	0.658
FACT-matched	20260827	41.3	37.8	37.4	37.4	0.0	0.668
FACT-generous	20260825	391.0	161.8	161.1	161.1	0.3	0.782
FACT-generous	20260826	747.1	508.8	488.3	488.3	0.0	0.768
FACT-generous	20260827	363.9	219.4	215.5	215.5	0.0	0.771
FORGE Transformer	20260825	952.1	951.2	872.4	872.4	3.3	0.656
FORGE Transformer	20260826	990.2	988.6	875.0	875.0	1.0	0.656
FORGE Transformer	20260827	952.5	951.8	840.5	840.5	0.0	0.618

Table 13: **Reaction decomposition outcomes under the common Ugi verifier.** Verified decomposition coverage is conditioned on molecular validity. Replay of every tuple in the returned exact set is a verifier invariant, not an estimate of reverse-search precision. Abstention and exact ambiguity retain their prespecified denominators. N/E denotes a conditional metric with no eligible valid product or returned decomposition.

Method	Valid products	Verified coverage	Replay invariant	Abstention	Exact ambiguity
RGFN	0.0 $\pm$ 0.0	N/E	N/E	N/E	N/E
DeFoG unconditional	1064.3 $\pm$ 129.9	81.2 $\pm$ 2.4	100.0 $\pm$ 0.0	18.8 $\pm$ 2.4	0.0 $\pm$ 0.0
GenMol/SAFE	1927.7 $\pm$ 19.0	0.0 $\pm$ 0.0	N/E	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Learned inventory selector	3072.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Finite catalogue oracle	3072.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Shared-null + post-hoc	598.0 $\pm$ 47.9	58.8 $\pm$ 3.5	100.0 $\pm$ 0.0	41.2 $\pm$ 3.5	0.0 $\pm$ 0.0
FACT-matched	186.7 $\pm$ 62.1	86.3 $\pm$ 4.4	100.0 $\pm$ 0.0	13.7 $\pm$ 4.4	0.0 $\pm$ 0.0
FACT-generous	1538.0 $\pm$ 656.9	56.6 $\pm$ 13.7	100.0 $\pm$ 0.0	43.4 $\pm$ 13.7	0.0 $\pm$ 0.0
FORGE Transformer	2964.3 $\pm$ 67.3	99.9 $\pm$ 0.0	100.0 $\pm$ 0.0	0.1 $\pm$ 0.0	0.0 $\pm$ 0.0

E Theoretical properties

This appendix establishes the formal properties of FORGE. We prove correctness and bounded completeness of the sparse representation, characterize finite-component reachability, prove fixed-coordinate invariance and Bayes optimality of the primary coordinatewise denoising loss, connect those posterior marginals to the local continuous-time rate, prove ideal population recovery of the terminal program-conditioned graph law, quantify matched-budget post-hoc rejection, and state termination plus the fail-closed route implementation invariant.

These results concern representation, population objectives, and computational verification. They do not imply that finite-data training assigns positive probability to every representable graph, that registered transformations cover arbitrary chemistry, that bounded route search is complete, or that a returned dossier guarantees experimental synthesis success.

E.1 Notation and setup

We use the following notation consistently.

Symbol	Meaning
$\Sigma_{\text{atom}}, \Sigma_{\text{bond}}$	atom-state and bond-state alphabets
N, K_{cl}	maximum active nodes and closure-edge budget
L_{samp}	finite Euler-step count
$\mathcal{I}$	six categorical state families
$\mathcal{F}(P), \mathcal{V}(P)$	fixed and variable coordinate sets
$\mathcal{C}_P$	admissible precursor-program inputs
$\mathfrak{B}_{\text{route}}$	frozen route-assessment contract
$\mathfrak{D}, \mathcal{D}_{\text{out}}$	dossier space and attempt-indexed dossier sequence
Q	proposal law in the post-hoc comparison

A molecular graph is $G = (V, E, \ell_V, \ell_E)$, where the underlying graph is finite, connected, undirected and simple, $\ell_V : V \rightarrow \Sigma_{\text{atom}}$, and $\ell_E : E \rightarrow \Sigma_{\text{bond}}$. Bond multiplicity belongs to ℓ_E ; parallel edges are not permitted. Define the set of label-preserving isomorphism classes

$$\mathfrak{G}_{N, K_{\text{cl}}} = \{[G]_{\cong} : |V| \leq N, \beta(G) = |E| - |V| + 1 \leq K_{\text{cl}}, \ell_V(V) \subseteq \Sigma_{\text{atom}}, \ell_E(E) \subseteq \Sigma_{\text{bond}}\}. \quad (46)$$

Let $\overline{\mathfrak{X}}_{N, K_{\text{cl}}}$ be the full padded categorical product space. A state belongs to its well-formed subset $\mathfrak{X}_{N, K_{\text{cl}}}$ exactly when:

1. for some $1 \leq n \leq N$, the active nodes are exactly $\{1, \dots, n\}$, every active node has a label in Σ_{atom} , and inactive node slots carry the null state;
2. node 1 is the unique root and carries the null parent-edge sentinel;
3. every $i > 1$ has $\pi_i \in \{1, \dots, i-1\}$ and a parent-edge label in Σ_{bond} ;
4. every active closure slot joins two distinct endpoints in $\{1, \dots, n\}$, carries a label in Σ_{bond} , and duplicates neither a tree edge nor another closure edge;
5. active closure triples are ordered lexicographically by endpoint pair; and
6. at most K_{cl} closure slots are active and every unused slot carries designated null endpoint and bond states.

The raw decoder returns a typed failure on every malformed state:

$$\text{Dec} : \overline{\mathfrak{X}}_{N, K_{\text{cl}}} \longrightarrow \mathfrak{G}_{N, K_{\text{cl}}} \cup \{\perp_{\text{dec}}\}. \quad (47)$$

E.2 Sparse graph representation

Lemma E.1 (Tree residual count). *Let $G = (V, E)$ be a connected finite graph and let T be any spanning tree of G . Then*

$$|E \setminus E(T)| = |E| - |V| + 1 = \beta(G). \quad (48)$$

872 *Proof.* A spanning tree on $|V|$ vertices contains exactly $|V| - 1$ edges. Because $E(T) \subseteq E$,

$$|E \setminus E(T)| = |E| - |E(T)| = |E| - (|V| - 1) = |E| - |V| + 1. \quad (49)$$

873 The final expression is the cyclomatic number of a connected graph. $\square$

Theorem E.2 (Correctness and bounded completeness of the sparse representation). *The decoder*

$$\text{Dec} : \mathfrak{X}_{N, K_{\text{cl}}} \longrightarrow \mathfrak{G}_{N, K_{\text{cl}}} \quad (50)$$

is well defined and surjective up to label-preserving graph isomorphism:

1. every $X \in \mathfrak{X}_{N, K_{\text{cl}}}$ decodes to a graph in $\mathfrak{G}_{N, K_{\text{cl}}}$; and
2. for every $G \in \mathfrak{G}_{N, K_{\text{cl}}}$, there exists $X \in \mathfrak{X}_{N, K_{\text{cl}}}$ such that $\text{Dec}(X) \cong G$.

874

875 *Proof.* Let $G = (V, E, \ell_V, \ell_E) \in \mathfrak{G}_{N, K_{\text{cl}}}$, and choose a spanning tree $T = (V, E_T)$ with $E_T \subseteq E$.
 876 Choose a root of T and enumerate its vertices $v_1, \dots, v_n$ in any root-first traversal. Every non-root
 877 vertex then appears after its parent.

878 Assign node i the label of v_i . For each $i > 1$, let $\pi_i < i$ be the index of the parent of v_i in T , and
 879 assign the corresponding tree-edge label. By Lemma E.1, the residual edge set $E \setminus E_T$ contains
 880 $\beta(G) \leq K_{\text{cl}}$ edges. Orient each residual edge by increasing endpoint index, attach its original
 881 edge label, and sort the resulting triples lexicographically by endpoint pair. The resulting state is an
 882 element of $\mathfrak{X}_{N, K_{\text{cl}}}$.

883 The decoder reconstructs every edge in E_T from the parent pointers and every edge in $E \setminus E_T$
 884 from the closure triples. It also preserves all node and edge labels. The map $i \mapsto v_i$ is therefore a
 885 label-preserving isomorphism from $\text{Dec}(X)$ to G .

886 Conversely, let $X \in \mathfrak{X}_{N, K_{\text{cl}}}$ have n active nodes. Since $\pi_i < i$, repeatedly following parent pointers
 887 from any node strictly decreases its index and must terminate at the root. The $n - 1$ parent edges
 888 therefore form a connected acyclic graph, hence a spanning tree. If the state contains $k \leq K_{\text{cl}}$ closure
 889 edges, the decoded graph has

$$|E| = (n - 1) + k \quad (51)$$

890 distinct edges. Its cyclomatic number is consequently

$$|E| - |V| + 1 = (n - 1 + k) - n + 1 = k \leq K_{\text{cl}}. \quad (52)$$

891 The remaining well-formedness conditions preserve the declared node and edge vocabularies and
 892 exclude loops or duplicate edges. Thus $\text{Dec}(X) \in \mathfrak{G}_{N, K_{\text{cl}}}$. $\square$

893 Global completeness does not by itself imply compatibility with a program face Ω_P . FORGE
 894 therefore restricts the morphology prior to

$$\mathcal{P}_{\text{adm}} = \{P : \mathfrak{X}_{N, K_{\text{cl}}} \cap \Omega_P \neq \emptyset\} \quad (53)$$

895 and treats $\text{Enc}_P(G, \alpha) \in \mathfrak{X}_{N, K_{\text{cl}}} \cap \Omega_P$ as an adapter-level implementation invariant checked for
 896 every admitted corpus record.

897 E.3 Finite-component reachability

Proposition E.3 (Finite-component reachability). *Assume that $\mathfrak{C}_P(\mathcal{C})$ is finite and that every forward-outcome set is finite. Then*

$$|\mathcal{G}_{\text{cat}}(P, \mathcal{C})| \leq \sum_{\mathbf{C} \in \mathfrak{C}_P(\mathcal{C})} |\mathcal{F}_P(\mathbf{C})|. \quad (54)$$

If $\mathfrak{C}_P(\mathcal{C}) \neq \emptyset$ and

$$m_{\text{max}, P}(\mathcal{C}) = \max_{\mathbf{C} \in \mathfrak{C}_P(\mathcal{C})} |\mathcal{F}_P(\mathbf{C})|, \quad (55)$$

then

$$|\mathcal{G}_{\text{cat}}(P, \mathcal{C})| \leq m_{\text{max}, P}(\mathcal{C}) |\mathfrak{C}_P(\mathcal{C})|. \quad (56)$$

898

Moreover,

$$G \in \mathcal{G}_{\text{cat}}(P, \mathcal{C}) \iff \mathcal{E}_P^*(G) \cap \mathfrak{C}_P(\mathcal{C}) \neq \emptyset. \quad (57)$$

Proof. The reachable set is a finite union of forward-outcome sets. The cardinality of a union is at most the sum of the cardinalities of its members, which proves the first inequality. Bounding each summand by $m_{\max, P}(\mathcal{C})$ proves the second.

By definition, G belongs to the union precisely when at least one $\mathbf{C} \in \mathfrak{C}_P(\mathcal{C})$ satisfies $G \in \mathcal{F}_P(\mathbf{C})$. This is equivalent to membership in $\mathcal{E}_P^*(G) \cap \mathfrak{C}_P(\mathcal{C})$. $\square$

Corollary E.4 (Catalogue assemblers cannot strongly escape). *For every $G \in \mathcal{G}_{\text{cat}}(P, \mathcal{C})$, $\text{Escape}^*(G; P, \mathcal{C}) = 0$. Consequently any assembler whose outputs are contained in $\mathcal{G}_{\text{cat}}(P, \mathcal{C})$ has zero strong-catalogue-escape yield per attempt.*

Proof. Membership in $\mathcal{G}_{\text{cat}}(P, \mathcal{C})$ supplies a catalogue tuple in $\mathcal{E}_P^*(G) \cap \mathfrak{C}_P(\mathcal{C})$, contradicting the second conjunct in the definition of strong escape. $\square$

E.4 Program-masked discrete flow

For a program P , let $\mathcal{I}(P)$ index all categorical coordinates, with coordinate alphabet $\mathcal{X}_i$. Partition it into fixed coordinates $\mathcal{F}(P)$ and variable coordinates $\mathcal{V}(P)$. Let $x_{P,i}$ be the prescribed value for $i \in \mathcal{F}(P)$ and define

$$\Omega_P = \{x : x_i = x_{P,i} \text{ for every } i \in \mathcal{F}(P)\}. \quad (58)$$

The projection Π_P restores $x_{P,i}$ on $\mathcal{F}(P)$ and is the identity on $\mathcal{V}(P)$.

The program-compatible source factorizes as

$$q_0^P(x) = \mathbf{1}[x_{\mathcal{F}(P)} = x_{P,\mathcal{F}(P)}] \prod_{i \in \mathcal{V}(P)} q_{0,i}^P(x_i), \quad q_{0,i}^P(a) > 0 \quad \forall a \in \mathcal{X}_i. \quad (59)$$

For a clean endpoint $x_1 \in \Omega_P$, the independent-coordinate conditional path is

$$q_{t|1}^P(x | x_1) = \mathbf{1}[x_{\mathcal{F}(P)} = x_{1,\mathcal{F}(P)}] \prod_{i \in \mathcal{V}(P)} [(1-t)q_{0,i}^P(x_i) + t\mathbf{1}\{x_i = x_{1,i}\}]. \quad (60)$$

Thus each coordinate is linearly interpolated conditional on the endpoint, while the joint law is their product. It is not the joint mixture $(1-t)q_0^P + tq_1^P$.

For coordinate i , write $p_{t,i}(v | a) = (1-t)q_{0,i}^P(v) + t\mathbf{1}\{v = a\}$, $\dot{p}_{t,i}(v | a) = \mathbf{1}\{v = a\} - q_{0,i}^P(v)$ and $S_{t,i}(a) = \{v : p_{t,i}(v | a) > 0\}$. The implementation's minimum univariate conditional rate is, for $u \neq v$,

$$R_{t,i}^*(u, v | a) = \frac{[\dot{p}_{t,i}(v | a) - \dot{p}_{t,i}(u | a)]_+ \mathbf{1}\{v \in S_{t,i}(a)\}}{|S_{t,i}(a)| p_{t,i}(u | a)}, \quad (61)$$

$$R_{t,i}^*(u, u | a) = - \sum_{v \neq u} R_{t,i}^*(u, v | a). \quad (62)$$

Full source support makes the off-diagonal denominator positive for the integration times used by the sampler. For $x \neq y$, define the local joint conditional generator

$$R_t^*(x, y | x_1, P) = \sum_{i \in \mathcal{V}(P)} \mathbf{1}\{y_{-i} = x_{-i}\} R_{t,i}^*(x_i, y_i | x_{1,i}). \quad (63)$$

It permits at most one coordinate to jump at an infinitesimal time. The denoiser induces the ideal marginal coordinate rate

$$R_{t,i}^\theta(u, v | x, P) = \sum_{a \in \mathcal{X}_i} p_{\theta,i}(a | x, t, P) R_{t,i}^*(u, v | a), \quad (64)$$

and hence, off diagonal,

$$R_t^\theta(x, y | P) = \sum_{i \in \mathcal{V}(P)} \mathbf{1}\{y_{-i} = x_{-i}\} R_{t,i}^\theta(x_i, y_i | x, P), \quad R_t^\theta(x, x | P) = - \sum_{y \neq x} R_t^\theta(x, y | P). \quad (65)$$

Fixed coordinates have identically zero off-diagonal rates.

For a grid $0 = t_0 < \dots < t_{L_{\text{samp}}} = 1$ and $\Delta t_k = t_{k+1} - t_k$, an uncapped independent-coordinate Euler kernel would use

$$K_{k,i}^\theta(v | x) = \Delta t_k R_{t_k,i}^\theta(x_i, v | x, P), \quad v \neq x_i, \quad (66)$$

$$K_{k,i}^\theta(x_i | x) = 1 - \Delta t_k \sum_{v \neq x_i} R_{t_k,i}^\theta(x_i, v | x, P), \quad (67)$$

and $\tilde{K}_k^\theta(y | x, P) = \mathbf{1}[y_{\mathcal{F}(P)} = x_{\mathcal{F}(P)}] \prod_{i \in \mathcal{V}(P)} K_{k,i}^\theta(y_i | x)$. These probabilities are nonnegative when $\Delta t_k \sum_{v \neq x_i} R_{t_k,i}^\theta(x_i, v | x, P) \leq 1$ for every coordinate and context.

The actual implementation uses one Monte Carlo clean endpoint $A_{k,i} \sim p_{\theta,i}(\cdot | X_k, t_k, P)$ per coordinate. It substitutes $R_{t_k,i}^*(\cdot, \cdot | A_{k,i})$ and multiplies every off-diagonal mass by

$$\alpha_{k,i} = \min \left\{ 1, \frac{0.999}{\Delta t_k \sum_{v \neq X_{k,i}} R_{t_k,i}^*(X_{k,i}, v | A_{k,i})} \right\}, \quad (68)$$

with $\alpha_{k,i} = 1$ when the denominator is zero. The diagonal receives the remaining mass. Coordinates are then drawn independently, $X_{k+1} \sim \prod_i \hat{K}_{k,i}$, followed by Π_P . This cap makes every finite-step kernel nonnegative and stochastic. It also means the code uses a finite-step Monte Carlo Euler approximation rather than exact Gillespie simulation of the ideal CTMC.

Proposition E.5 (Program-fixed invariance). *For every $P \in \mathcal{P}_{\text{adm}}$, corruption and numerical sampling satisfy $\Pr[X_t \in \Omega_P] = 1$ and $\Pr[X_k \in \Omega_P \text{ for all } k = 0, \dots, L_{\text{samp}}] = 1$.*

Proof. The source and conditional path assign zero mass outside the fixed face. Fixed coordinates also have zero off-diagonal rates. Finally, $\Pi_P(x) \in \Omega_P$ for every raw x , so the implementation safeguard preserves the claim by induction over finite steps. $\square$

Let $X_1 = \text{Enc}_P(G, \alpha)$ and define the population law

$$\nu(dG, d\alpha, dP, dt, dX_t) = \mu_{\text{bal}}(dG, d\alpha, dP) \mathcal{U}(dt) q_{t|1}^P(dX_t | X_1). \quad (69)$$

For $i \in \mathcal{V}(P)$, write $\rho_i(a | X_t, t, P) = \Pr_\nu[X_{1,i} = a | X_t, t, P]$.

Proposition E.6 (Bayes optimality of the primary coordinatewise masked-denoising loss). *Assume an unrestricted coordinatewise conditional model class and positive state-family weight. On the event $i \in \mathcal{V}(P)$, every population minimizer of the primary masked denoising term satisfies*

$$p_{\theta,i}^*(a | X_t, t, P) = \rho_i(a | X_t, t, P) \quad (70)$$

for every $a \in \mathcal{X}_i$, ν -almost surely. This proposition does not characterize the optimizer produced by auxiliary losses, finite capacity or PCGrad.

Proof. Fix a variable coordinate and context. Its conditional cross-entropy is $H(\rho_i) + D_{\text{KL}}(\rho_i \| p_{\theta,i})$, which is minimized exactly at $p_{\theta,i} = \rho_i$. The positive family weight and positive normalization by the number of variable coordinates do not alter that pointwise minimizer. Fixed coordinates do not appear in the primary loss. $\square$

Corollary E.7 (Population recovery of the local rate). *Suppose the conditional joint generator decomposes coordinatewise as above. Substituting any population minimizer $p_{\theta,i}^*$ of the primary*

masked denoising term into the marginal-rate formula recovers the population local generator R_t almost surely on the training path.

948

949 *Proof.* The proposition identifies $p_{\theta,i}^*$ with the true clean-endpoint posterior marginal. The displayed
 950 rate is linear in that marginal, so substituting it performs the conditional expectation of $R_{t,i}^*$ required
 951 for each local coordinate generator. Summing the one-coordinate terms recovers the joint local
 952 generator. $\square$

Theorem E.8 (Ideal population correctness of program-conditioned generation). *Fix a program P with positive mass under μ_{bal} , and let*

$$q_1^P = \mathcal{L}(\text{Enc}_P(G, \alpha) \mid P), \quad \mu_G^P = \mathcal{L}([G]_{\cong} \mid P) \quad (71)$$

be the encoded endpoint law and corresponding conditional graph law. Assume finite coordinate alphabets and a full-support variable-coordinate source; every admitted endpoint decodes by exact round trip, is molecularly valid and satisfies $\widehat{\text{L1}}(G, P) = 1$; the unrestricted coordinatewise model is at a population minimum of the primary masked denoising term; and sampling starts from the program source and exactly simulates the induced local continuous-time Markov chain. Then, writing $G_1 = \text{Dec}(X_1)$, the ideal terminal state satisfies

$$\mathcal{L}(X_1 \mid P) = q_1^P, \quad \text{Dec}_{\#} q_1^P = \mu_G^P, \quad (72)$$

$$\Pr[\text{Valid}(G_1) = 1, \widehat{\text{L1}}(G_1, P) = 1 \mid P] = 1. \quad (73)$$

953

954 **Formal path and proof of Theorem E.8.** For the fixed program in the theorem, write the encoded
 955 endpoint mass function and population path as

$$q_1^P(x) = \Pr_{\mu_{\text{bal}}}[\text{Enc}_P(G, \alpha) = x \mid P], \quad (74)$$

$$q_t^P(x) = \sum_{x_1} q_1^P(x_1) q_{t|1}^P(x \mid x_1). \quad (75)$$

956 The theorem's exact-simulation assumption uses the displayed rates $R_{t,i}^*$ on $0 \leq t < 1$, initial law
 957 $X_0 \sim q_0^P$ and terminal law defined by $t \uparrow 1$. We prove the stronger path identity

$$\mathcal{L}(X_t \mid P) = q_t^P \quad (0 \leq t < 1), \quad \mathcal{L}(X_1 \mid P) = q_1^P. \quad (76)$$

958 For clarity, its conditional graph mass function is

$$\mu_G^P([G]_{\cong}) = \Pr_{\mu_{\text{bal}}}[[G']_{\cong} = [G]_{\cong} \mid P], \quad (77)$$

959 and the remaining conclusions are

$$\text{Dec}_{\#} q_1^P = \mu_G^P \quad (78)$$

960 on $\mathfrak{G}_{N, K_{\text{cl}}} \cup \{\perp_{\text{dec}}\}$, with zero mass at $\perp_{\text{dec}}$, and

$$G_1 := \text{Dec}(X_1), \quad \Pr[G_1 \in \mathfrak{G}_{N, K_{\text{cl}}}, \text{Valid}(G_1) = 1, \widehat{\text{L1}}(G_1, P) = 1 \mid P] = 1. \quad (79)$$

961 *Proof of Theorem E.8.* Fix an endpoint x_1 . For one coordinate, let $p_t(v) = p_{t,i}(v \mid x_{1,i})$ and
 962 $\dot{p}_t(v) = \dot{p}_{t,i}(v \mid x_{1,i})$. The off-diagonal definition of $R_{t,i}^*$ gives the net probability flux into state v as

$$\sum_u p_t(u) R_{t,i}^*(u, v \mid x_{1,i}) = \frac{1}{|S_{t,i}|} \sum_{u \in S_{t,i}} ([\dot{p}_t(v) - \dot{p}_t(u)]_+ - [\dot{p}_t(u) - \dot{p}_t(v)]_+) \quad (80)$$

$$= \dot{p}_t(v) - \frac{1}{|S_{t,i}|} \sum_{u \in S_{t,i}} \dot{p}_t(u) = \dot{p}_t(v), \quad (81)$$

because the derivative of a probability vector sums to zero. Thus each univariate rate realizes its coordinate path. The sum of one-coordinate generators realizes the product conditional path $q_{t|1}^P$ by the product rule:

$$\partial_t q_{t|1}^P(y | x_1) = \sum_x q_{t|1}^P(x | x_1) R_t^*(x, y | x_1, P). \quad (82)$$

Averaging this identity over q_1^P and applying Bayes' rule yields

$$\partial_t q_t^P(y) = \sum_x q_t^P(x) \sum_{x_1} \Pr[X_1 = x_1 | X_t = x, P] R_t^*(x, y | x_1, P) \quad (83)$$

$$= \sum_x q_t^P(x) R_t(x, y | P). \quad (84)$$

By Corollary E.7, $R_t^{\theta^*} = R_t$ almost surely on the population path. On every compact interval $[0, 1 - \eta]$, $\eta > 0$, this finite-state forward equation has a unique solution with initial law q_0^P , hence the exactly simulated CTMC has marginal q_t^P . Since each conditional coordinate path converges to its endpoint point mass, $q_t^P \rightarrow q_1^P$ as $t \uparrow 1$.

Finally, deterministic decode round trip sends every encoded endpoint to its source graph. Summing q_1^P over all endpoint encodings of one graph class therefore gives μ_G^P . The assumed endpoint validity and exact replay make the final probability-one statement immediate. $\square$

Corollary E.9 (Decoder contraction and approximate failure bound). *Under the endpoint assumptions of Theorem E.8, let $\tilde{q}_1^P$ be any approximate terminal raw-state law and define*

$$\mathcal{B}_P = \{x : \text{Dec}(x) = \perp_{\text{dec}}\} \cup \left\{x : \text{Dec}(x) = G \in \mathfrak{G}_{N, K_{\text{cl}}}, \text{Valid}(G) \widehat{\text{L1}}(G, P) = 0\right\}. \quad (85)$$

Then

$$\|\text{Dec}_{\#} \tilde{q}_1^P - \mu_G^P\|_{\text{TV}} \leq \|\tilde{q}_1^P - q_1^P\|_{\text{TV}}, \quad (86)$$

and

$$\Pr_{X \sim \tilde{q}_1^P}[X \in \mathcal{B}_P] \leq \|\tilde{q}_1^P - q_1^P\|_{\text{TV}}. \quad (87)$$

974

Proof. Total variation cannot increase under a measurable deterministic map, which proves the first inequality. The failure event has probability zero under q_1^P by Theorem E.8; its probability under $\tilde{q}_1^P$ is therefore at most the total-variation distance between the two raw-state laws. $\square$

The theorem is intentionally an ideal population statement. It does not characterize the finite-capacity Transformer trained with auxiliary losses and PCGrad, does not bound the Monte Carlo endpoint approximation or capped Euler discretization, and does not imply support outside q_1^P or strong catalogue escape. Corollary E.9 identifies the single approximation quantity whose control would transfer ideal endpoint correctness to the implemented sampler.

983 E.5 Matched-budget post-hoc filtering

984 Fix a reaction program P , and define

$$\mathcal{Y} = \mathfrak{G}_{N, K_{\text{cl}}} \cup \mathfrak{F}_{\text{gen}}, \quad A_P(y) = \mathbf{1}\left[y \in \mathfrak{G}_{N, K_{\text{cl}}}, \text{Valid}(y) = 1, \widehat{\text{L1}}(y, P) = 1\right]. \quad (88)$$

985 Thus $A_P(y) = 0$ for invalid-decode and other generation failures. For a proposal law Q on $\mathcal{Y}$,
986 conditional on a fixed trained model and requested program, let

$$Z_Q(P) = \Pr_{G \sim Q}[A_P(G) = 1] = \mathbb{E}_{G \sim Q}[A_P(G)] \quad (89)$$

987 denote the exact-L1 acceptance probability.

Proposition E.10 (Exact law of matched-budget post-hoc filtering). *For B conditionally independent attempts $Y_1, \dots, Y_B \sim Q$, $N_B = \sum_{b=1}^B A_P(Y_b)$ satisfies*

$$N_B \sim \text{Binomial}(B, Z_Q(P)) \quad (90)$$

for every $Z_Q(P) \in [0, 1]$. If $Z_Q(P) > 0$, rejection filtering additionally produces

$$Q_{\text{post}}(g \mid P) = \Pr_{G \sim Q}[G = g \mid A_P(G) = 1] = \frac{Q(g)A_P(g)}{Z_Q(P)}. \quad (91)$$

and consequently

$$\mathbb{E}[N_B] = BZ_Q(P), \quad (92)$$

$$\text{Var}(N_B) = BZ_Q(P)(1 - Z_Q(P)), \quad (93)$$

$$\Pr[N_B = 0] = (1 - Z_Q(P))^B. \quad (94)$$

If T_m is the number of independent attempts required to obtain m accepted products, then

$$\mathbb{E}[T_m] = \frac{m}{Z_Q(P)}. \quad (95)$$

Finally, for a program-conditioned proposal Q_P and an unconditioned proposal Q_0 ,

$$\mathbb{E}[N_B^P] - \mathbb{E}[N_B^0] = B(Z_{Q_P}(P) - Z_{Q_0}(P)). \quad (96)$$

988

989 *Proof.* Conditional independence makes the acceptance indicators Bernoulli with common probability
990 $Z_Q(P)$, including the degenerate case $Z_Q(P) = 0$, so their sum is binomial. When $Z_Q(P) > 0$,
991 Bayes’ rule gives

$$\Pr[G = g \mid A_P(G) = 1] = \frac{\Pr[A_P(G) = 1 \mid G = g] \Pr[G = g]}{\Pr[A_P(G) = 1]} \quad (97)$$

$$= \frac{A_P(g)Q(g)}{Z_Q(P)}. \quad (98)$$

992 The waiting time T_m has the negative-binomial distribution with success probability $Z_Q(P)$, whose
993 expectation is $m/Z_Q(P)$. Applying the binomial expectation separately to Q_P and Q_0 and subtracting
994 proves the final identity. $\square$

995 The proposition compares sampling efficiency only. It does not imply that conditioning improves
996 $Z_Q(P)$; the matched production experiment tests that empirical claim. The binomial count is the
997 total accepted-attempt count, not the number of distinct accepted products.

998 E.6 Recursive route construction

Lemma E.11 (Termination of bounded route assessment). *Fix $\mathfrak{B}_{\text{route}}$ with finite maximum route depth $D_{\mathfrak{B}}$. Assume finitely many candidate root decompositions, finitely many alternative expansions per molecule, finitely many reactants per expansion, and termination of every evidence, proposal and forward-verification call. Then $\text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T)$ terminates in $\mathfrak{D} \cup \mathfrak{F}_{\mathfrak{B}}$.*

999

1000 *Proof.* We use induction on the remaining depth $d = D_{\mathfrak{B}} - k$ at a node of depth k . When $d = 0$, the
1001 assessor performs the terminating evidence query and either accepts a terminal node, returns its typed
1002 failure, or rejects an expansion with a depth-exhaustion state. Suppose the claim holds whenever
1003 fewer than $d > 0$ levels remain. At the current node, the finite sequence of terminating proposal,
1004 evidence and verification calls returns either a typed non-expansion outcome or one of finitely many
1005 finite reactant tuples. In the latter case, each child is assessed at depth $k + 1$ and therefore has $d - 1$
1006 levels remaining. Every child assessment terminates by the induction hypothesis, and a finite tuple
1007 of finite child trees forms a finite parent tree. The finite root-decomposition loop also terminates.
1008 Rejecting an ancestor identity additionally prevents a circular molecular dependency from being
1009 accepted, but finite depth alone establishes termination. $\square$

Implementation invariant E.12 (Fail-closed synthesis-dossier implementation). *For the frozen assessor implementation, a successful return $\text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T) = T \in \mathfrak{D}$ is admitted only after checking $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$. Every other branch returns a member of the typed failure set $\mathfrak{F}_{\mathfrak{B}}$. This is an implementation invariant, not a completeness theorem for the chemical search space.*

1010

1011 The invariant records four independently checked conditions: the dossier is a well-formed tree rooted
 1012 at G ; its selected root trace belongs to $\hat{\mathcal{E}}_P(G)$; every upstream reaction is evidence-admitted and
 1013 satisfies $\text{target}(r) \in \mathcal{F}_r(\text{reactants}(r))$; and every leaf closes under the dated availability policy. A
 1014 typed failure means only that no complete dossier was returned under the frozen contract, not that no
 1015 route exists under a larger search budget, a different admitted reaction set, or a later supplier snapshot.

1016 F Generated-structure atlas

1017 Table 14 reports 12 deterministic display examples from the same frozen seed-0 Ugi population used
 1018 in Figure 2. The display begins with the lowest constitutional SHA-256 rank and then applies ECFP4
 1019 MaxMin selection among the 23 eligible, constitutionally unique products. This promotes visual
 1020 coverage without using activity, route closure, procurement evidence or human preference. Each
 1021 row includes the canonical constitutional SMILES so that the displayed identity is explicit; the exact
 1022 strings are also stored in the hash-pinned atlas receipt.

Table 14: **FORGE generated-structure atlas**. The 12 rows form one multipage table, indexed by display order. Each constitutional graph is followed by its canonical SMILES. The decorated three-dimensional column shows one deterministic post-hoc RDKit ETKDGv3 structure optimized with up to 400 MMFF94s iterations. Purple denotes the amine-derived head, teal the aldehyde-derived body and tail, orange the isocyanide-derived tail and gray the assembly-introduced oxygen; these assignments are recovered through exact atom-mapped forward replay. FORGE does not generate three-dimensional coordinates, and these views are not predicted bioactive conformations. Exact L1 replay and descriptor-manifold membership are structural diagnostics, not evidence of activity, route closure or synthesis success. These are deterministic display examples, not experimental candidates.

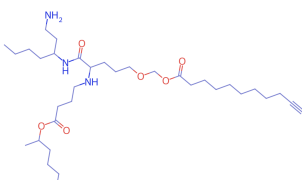
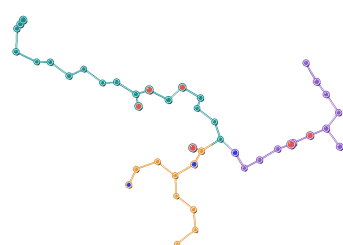
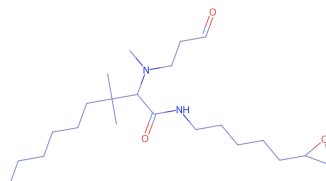
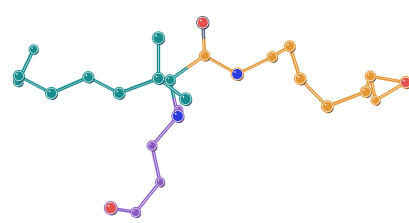
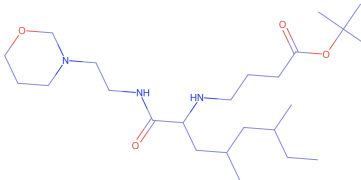
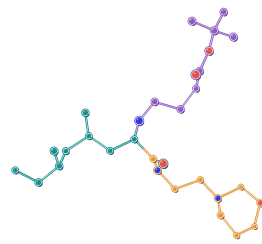
	CONSTITUTIONAL GRAPH AND CANONICAL SMILES	DECORATED 3D VIEW
1	 <chem>C#CCCCCCCCC(=O)OCOC(CCC(=O)OC(C)CCCC)C(=O)NC(CCN)CCCC</chem>	
2	 <chem>CCCCCCC(C)(C)C(C(=O)NCCCCC1CO1)N(C)CCC=O</chem>	
3	 <chem>CCC(C)CC(C)CC(NCCCC(=O)OC(C)(C)C)C(=O)NCCN1CCCCO1</chem>	

Table 14, continued

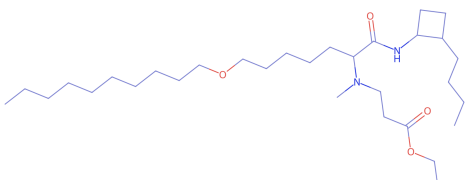
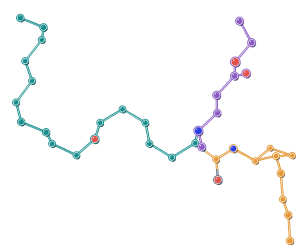
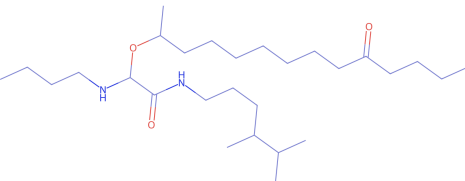
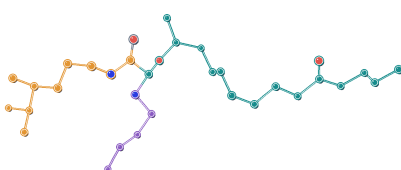
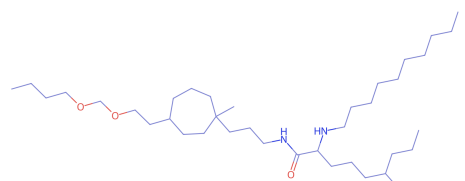
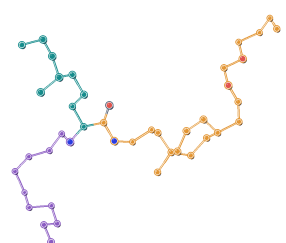
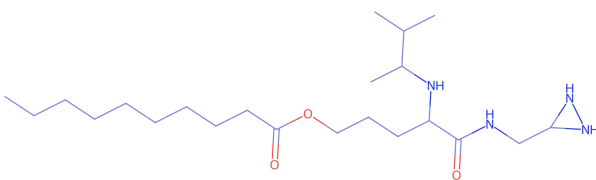
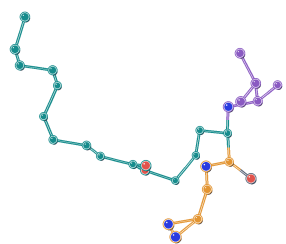
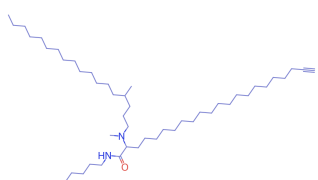
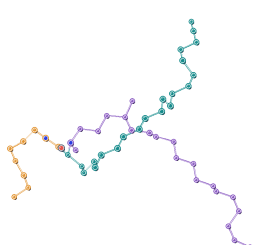
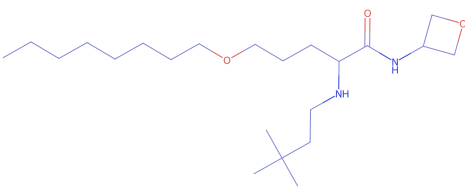
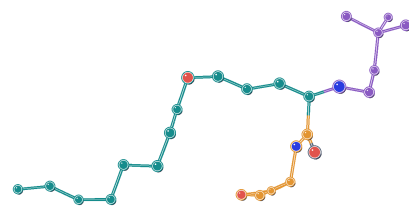
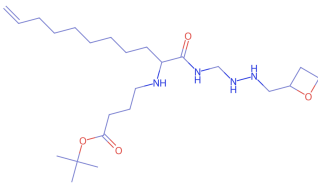
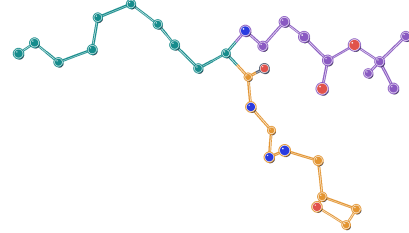
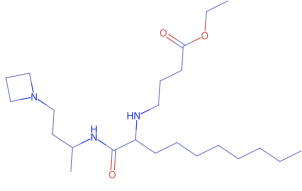
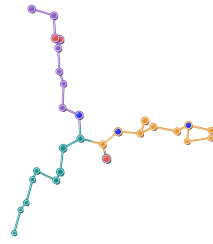
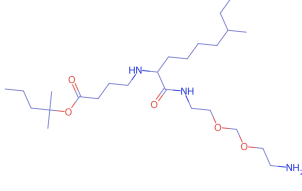
	CONSTITUTIONAL GRAPH AND CANONICAL SMILES	DECORATED 3D VIEW
4	 <chem>CCCCCCCCCOC(=O)N1CCCCC1N(C)CCC(=O)OCC</chem>	
5	 <chem>CCCCNC(OC(C)CCCCC(=O)CCCC)C(=O)NCCCC(C)C(C)C</chem>	
6	 <chem>CCCCCCCCCNC(CCCC(C)CCC)C(=O)NCCCC1(C)CCCC(COC(C)CCCC)CC1</chem>	
7	 <chem>CCCCCCCCC(=O)OCCCC(NC(C)C(C)C)C(=O)NCC1NN1</chem>	
8	 <chem>C#CCCCCCCCCCCCCCCCC(C(=O)NCCCCC)N(C)CCCC(C)CCCCCCCCCCCCC</chem>	

Table 14, continued

	CONSTITUTIONAL GRAPH AND CANONICAL SMILES	DECORATED 3D VIEW
9	 <chem>CCCCCCCCOCCCC(NCCC(C)(C)C)C(=O)NC1COC1</chem>	
10	 <chem>C=CCCCCCCC(NCCCC(=O)OC(C)(C)C)C(=O)NCNCCC1COC1</chem>	
11	 <chem>CCCCCCCC(NCCCC(=O)OCC)C(=O)NC(C)CCN1CCC1</chem>	
12	 <chem>CCCC(C)(C)OC(=O)CCCNC(CCCCC(C)CC)C(=O)NCCOC(C)CN</chem>	